Signals and Systems

Lecture 15: Discrete-Time Fourier Representation

(Continued)

Instructor: Prof. Ting Zhang
Zhejiang University

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Adapted from the materials provided on
the MIT OpenCourseWare

DT Fourier Series

DT Fourier series represent DT signals in terms of the amplitudes and phases of harmonic components.

$$x[n] = \sum a_k e^{jk\Omega_0 n}$$

The period N of all harmonic components is the same.

DT Fourier Series

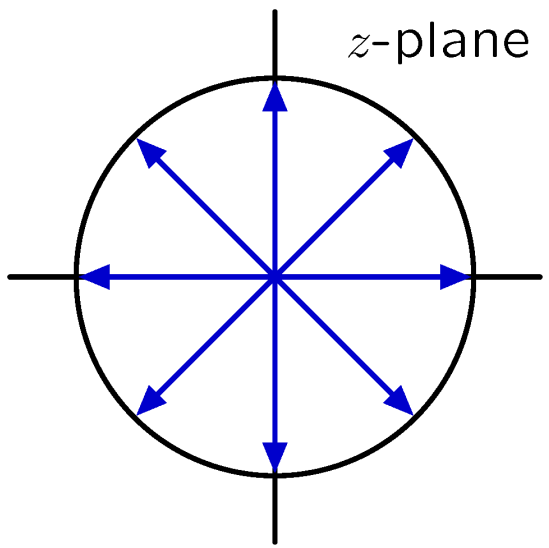
There are N distinct complex exponentials with period N.

If $e^{j\Omega n}$ is periodic in N then

$$e^{j\Omega n} = e^{j\Omega(n+N)} = e^{j\Omega n}e^{j\Omega N}$$

and $e^{j\Omega N}$ must be 1, and Ω must be one of the Nth roots of 1.

Example: $N = 8$

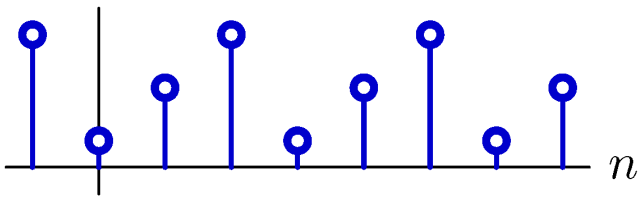


DT Fourier Series

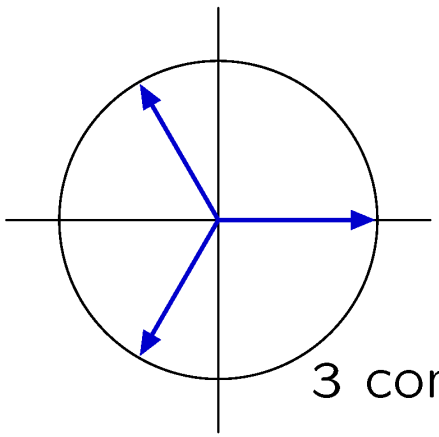
There are N distinct complex exponentials with period N.

These can be combined via Fourier series to produce periodic time signals with N independent samples.

Example: periodic in N=3

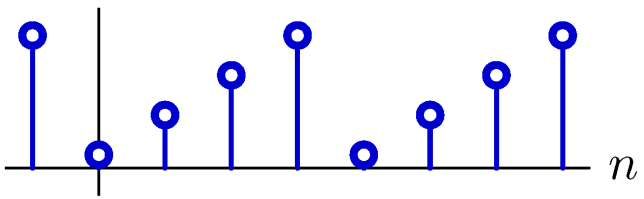


3 samples repeated in time

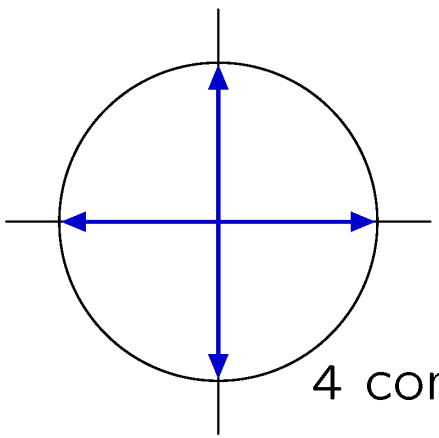


3 complex exponentials

Example: periodic in N=4



4 samples repeated in time



4 complex exponentials

DT Fourier Series

DT Fourier series represent DT signals in terms of the amplitudes and phases of harmonic components.

$$x[n] = x[n+N] = \sum_{k=0}^{N-1} a_k e^{jk\Omega_0 n} \quad ; \; \Omega_0 = \frac{2\pi}{N}$$

N equations (one for each point in time n) in N unknowns (a_k).

Example: $N = 4$

$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix} = \begin{bmatrix} e^{j\frac{2\pi}{N}0\cdot 0} & e^{j\frac{2\pi}{N}1\cdot 0} & e^{j\frac{2\pi}{N}2\cdot 0} & e^{j\frac{2\pi}{N}3\cdot 0} \\ e^{j\frac{2\pi}{N}0\cdot 1} & e^{j\frac{2\pi}{N}1\cdot 1} & e^{j\frac{2\pi}{N}2\cdot 1} & e^{j\frac{2\pi}{N}3\cdot 1} \\ e^{j\frac{2\pi}{N}0\cdot 2} & e^{j\frac{2\pi}{N}1\cdot 2} & e^{j\frac{2\pi}{N}2\cdot 2} & e^{j\frac{2\pi}{N}3\cdot 2} \\ e^{j\frac{2\pi}{N}0\cdot 3} & e^{j\frac{2\pi}{N}1\cdot 3} & e^{j\frac{2\pi}{N}2\cdot 3} & e^{j\frac{2\pi}{N}3\cdot 3} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

DT Fourier Series

DT Fourier series represent DT signals in terms of the amplitudes and phases of harmonic components.

$$x[n] = x[n+N] = \sum_{k=0}^{N-1} a_k e^{jk\Omega_0 n} \quad ; \; \Omega_0 = \frac{2\pi}{N}$$

N equations (one for each point in time n) in n unknowns (a_k).

Example: $N = 4$

$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

Orthogonality

Solving these equations is simple because these complex exponentials are orthogonal to each other.

$$\sum_{n=0}^{N-1} e^{j\Omega_0 kn} e^{-j\Omega_0 ln} = \sum_{n=0}^{N-1} e^{j\Omega_0 (k-l)n}$$

$$= \begin{cases} N & ;\ k = l \\ \frac{1-e^{j\Omega_0 (k-l)N}}{1-e^{j\Omega_0 (k-l)}} = 0 & ;\ k \neq l \end{cases}$$

$$= N\delta[k-l]$$

Orthogonality

We can use the orthogonality property of these complex exponentials to sift out the Fourier series coefficients, one at a time.

Assume $x[n] = \sum_{k=0}^{N-1} a_k e^{jk\Omega_0 n}$

Multiply both sides by the complex conjugate of the lth harmonic, and sum over time.

$$\sum_{n=0}^{N-1} x[n] e^{-jl\Omega_0 n} = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} a_k e^{jk\Omega_0 n} e^{-jl\Omega_0 n} = \sum_{k=0}^{N-1} a_k \sum_{n=0}^{N-1} e^{jk\Omega_0 n} e^{-jl\Omega_0 n}$$

$$= \sum_{k=0}^{N-1} a_k N \delta[k-l] = N a_l$$

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\Omega_0 n}$$

DT Fourier Series

Since both $x[n]$ and a_k are periodic in N, the sums can be taken over any N successive indices.

Notation. If $f[n]$ is periodic in N, then

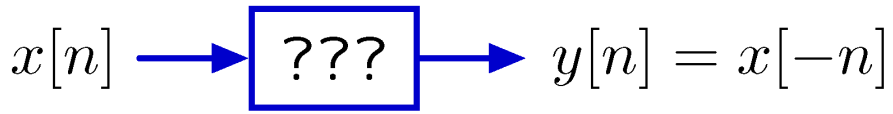
$$\sum_{n=0}^{N-1} f[n] = \sum_{n=1}^{N} f[n] = \sum_{n=2}^{N+1} f[n] = \cdots = \sum_{n=<N>} f[n]$$

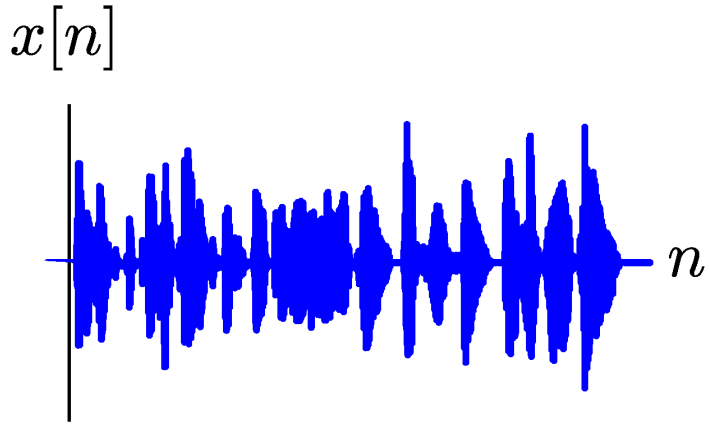
DT Fourier Series

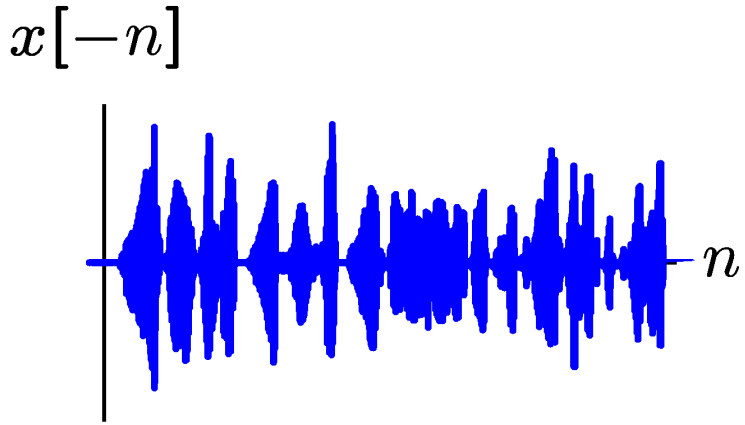
$$a_k = a_{k+N} = \frac{1}{N} \sum_{n=<N>} x[n] e^{-j\Omega_0 n} \quad ; \quad \Omega_0 = \frac{2\pi}{N} \qquad \text{("analysis" equation)}$$

$$x[n] = x[n+N] = \sum_{k=<N>} a_k e^{jk\Omega_0 n} \qquad \text{("synthesis" equation)}$$

Check Yourself

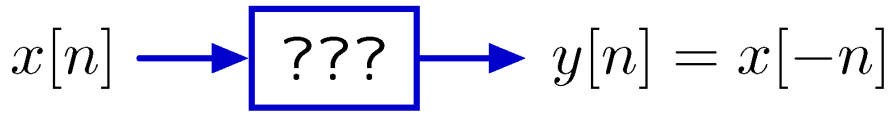


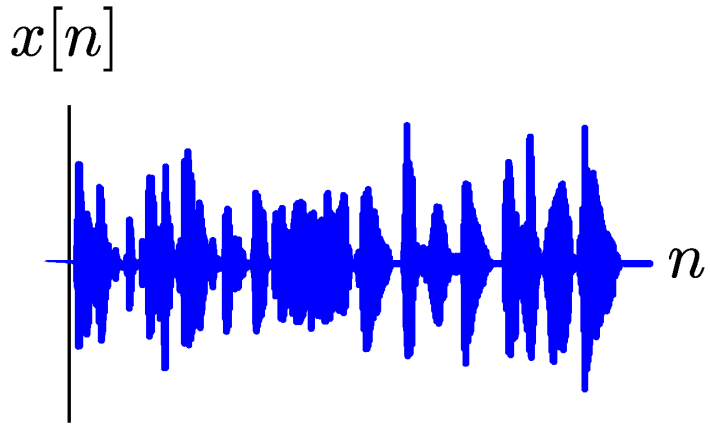


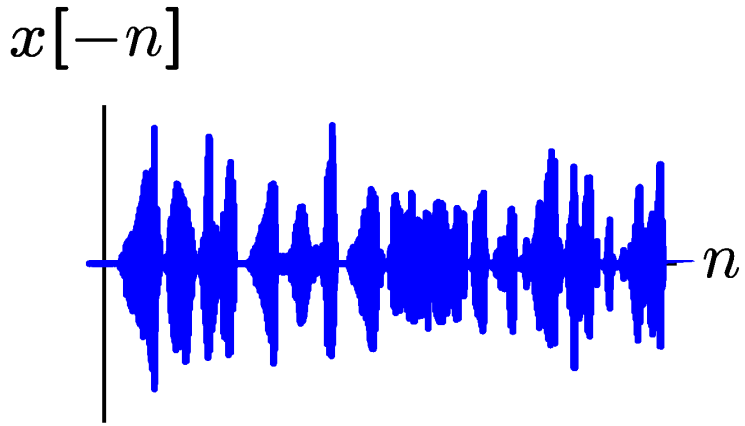


`artificial speech synthesized by Robert Donovan`

Check Yourself







How are the phases of X and Y related?

Check Yourself

$$a_k = \frac{1}{N} \sum_n x[n] e^{-jk\Omega_0 n}$$

$$b_k = \frac{1}{N} \sum_n x[-n] e^{-jk\Omega_0 n} = \frac{1}{N} \sum_m x[m] e^{jk\Omega_0 m} = a_{-k}$$

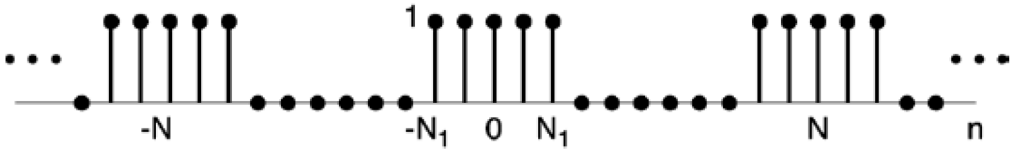
$$b_k = a_{-k} = a_k^* = |a_k| e^{-j\angle a_k}$$

Flipping $x[n]$ about $n = 0$ flips a_k about $k = 0$.
Since $x[n]$ is real-valued, a_k is conjugate symmetric: $a_{-k} = a_k^*$.
The angles are negated at all frequencies.

Check Yourself

Ex.#5 Determine the discrete Fourier series representation for

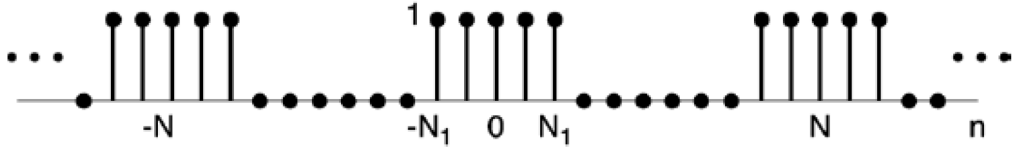
DT Square Wave



Check Yourself

Ex.#5 Determine the discrete Fourier series representation for

DT Square Wave



$$a_0 = \frac{1}{N}\sum_{n=-N_1}^{N_1} x[n] = \frac{(2N_1+1)}{N} = a_N = a_{-N} = a_{6N} = \cdots$$

For $k \neq$ multiple of N:

$$a_k = \frac{1}{N}\sum_{n=-N_1}^{N_1} e^{-jk\omega_0 n} \overset{n=m-N_1}{=} \frac{1}{N}\sum_{m=0}^{2N_1} e^{-jk\omega_0 (m-N_1)}$$

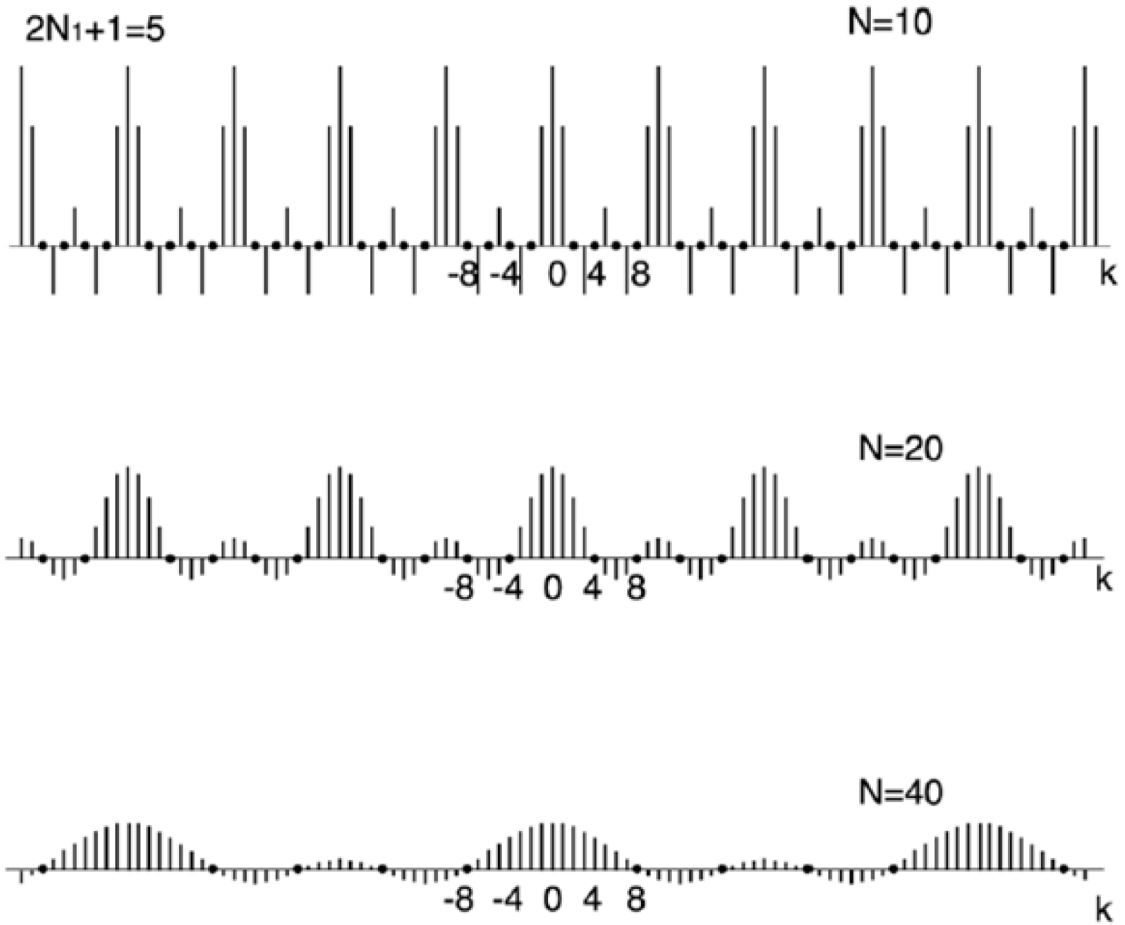
$$= \frac{1}{N} e^{jk\omega_0 N_1} \sum_{m=0}^{2N_1} (e^{-jk\omega_0})^m = \frac{1}{N} e^{jk\omega_0 N_1} \frac{1-e^{-jk\omega_0(2N_1+1)}}{1-e^{jk\omega_0}}$$

$$= \frac{1}{N} \frac{\sin[k(N_1+1/2)\omega_0]}{\sin(k\omega_0/2)} = \frac{1}{N} \frac{\sin[2\pi k(N_1+1/2)/N]}{\sin(\pi k/N)}$$

Check Yourself

Ex.#5 Determine the discrete Fourier series representation for

$$a_k = \frac{1}{N}\frac{\sin\left[2\pi k(N_1 + 1/2)/N\right]}{\sin(\pi k/N)}$$



Scaling

DT Fourier series are important computational tools.

However, the DT Fourier series do not scale well with the length N.

$$a_k = a_{k+2} = \frac{1}{2}\sum_{n=<2>} x[n]e^{-jk\Omega_0 n} = \frac{1}{2}\sum_{n=<2>} e^{-jk\frac{2\pi}{2}n} = \frac{1}{2}\sum_{n=<2>} x[n](-1)^{-k}$$

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

$$a_k = a_{k+4} = \frac{1}{4}\sum_{n=<4>} x[n]e^{-jk\Omega_0 n} = \frac{1}{4}\sum_{n=<4>} e^{-jk\frac{2\pi}{4}n} = \frac{1}{4}\sum_{n=<4>} x[n]j^{-kn}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

Number of multiples increases as N^2.

Fast Fourier "Transform"

Exploit structure of Fourier series to simplify its calculation.

Divide FS of length $2N$ into two of length N (divide and conquer).

Matrix formulation of 8-point FS:

$$\begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{bmatrix} = \begin{bmatrix} W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 \\ W_8^0 & W_8^1 & W_8^2 & W_8^3 & W_8^4 & W_8^5 & W_8^6 & W_8^7 \\ W_8^0 & W_8^2 & W_8^4 & W_8^6 & W_8^0 & W_8^2 & W_8^4 & W_8^6 \\ W_8^0 & W_8^3 & W_8^6 & W_8^1 & W_8^4 & W_8^7 & W_8^2 & W_8^5 \\ W_8^0 & W_8^4 & W_8^0 & W_8^4 & W_8^0 & W_8^4 & W_8^0 & W_8^4 \\ W_8^0 & W_8^5 & W_8^2 & W_8^7 & W_8^4 & W_8^1 & W_8^6 & W_8^3 \\ W_8^0 & W_8^6 & W_8^4 & W_8^2 & W_8^0 & W_8^6 & W_8^4 & W_8^2 \\ W_8^0 & W_8^7 & W_8^6 & W_8^5 & W_8^4 & W_8^3 & W_8^2 & W_8^1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \\ x[4] \\ x[5] \\ x[6] \\ x[7] \end{bmatrix}$$

where $W_N = e^{-j\frac{2\pi}{N}}$

$8 \times 8 = 64$ multiplications

Similarities and differences between CT and DT

Similarities and differences between CT and DT.

DT frequency response

- vector diagrams (similar to CT)
- frequency response on unit circle in z-plane ($j\omega$ axis in CT)

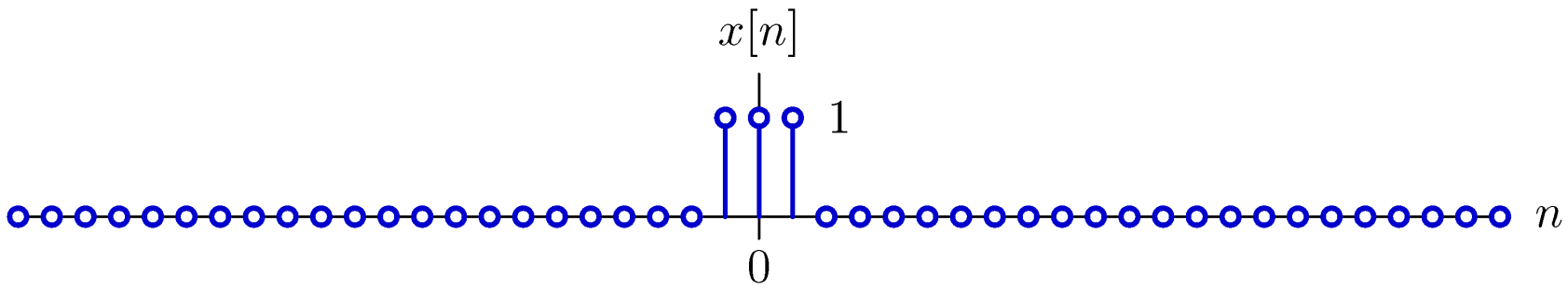
DT Fourier series

- represent signal as sum of harmonics (similar to CT)
- finite number of periodic harmonics (unlike CT)
- finite sum (unlike CT)

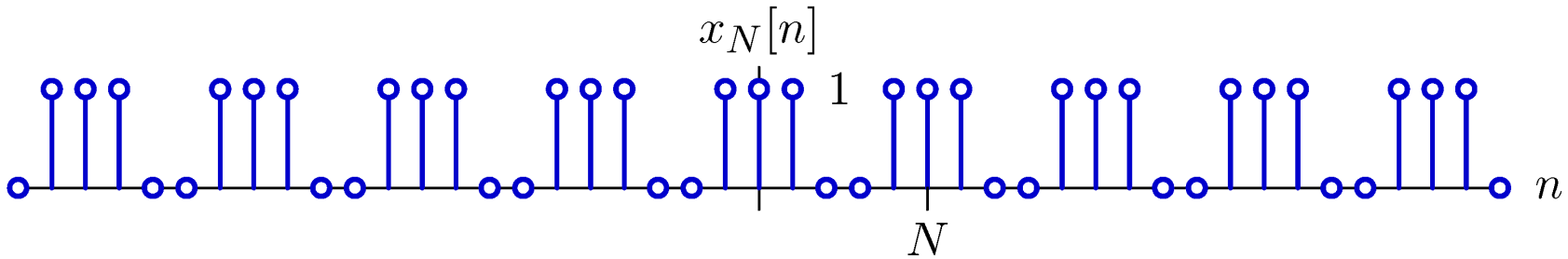
Fourier Transform: Generalize to Aperiodic Signals

An aperiodic signal can be thought of as periodic with infinite period.

Let $x[n]$ represent an aperiodic signal DT signal.



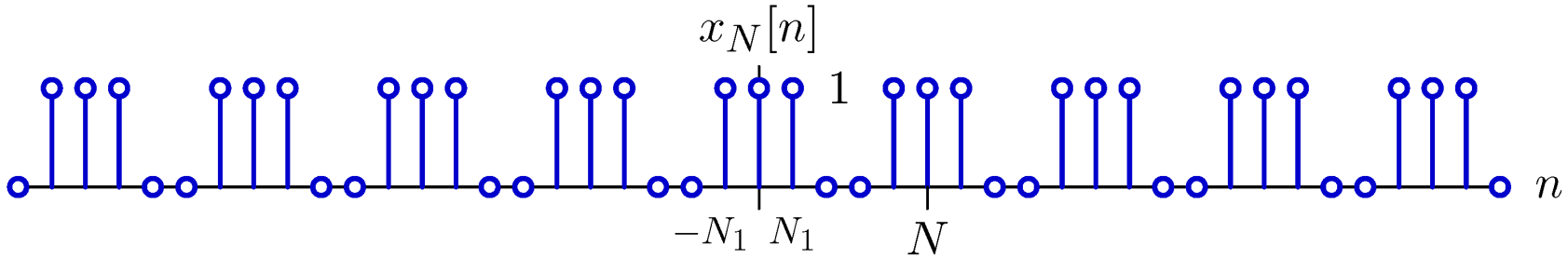
"Periodic extension": $x_N[n] = \sum_{k=-\infty}^{\infty} x[n+kN]$



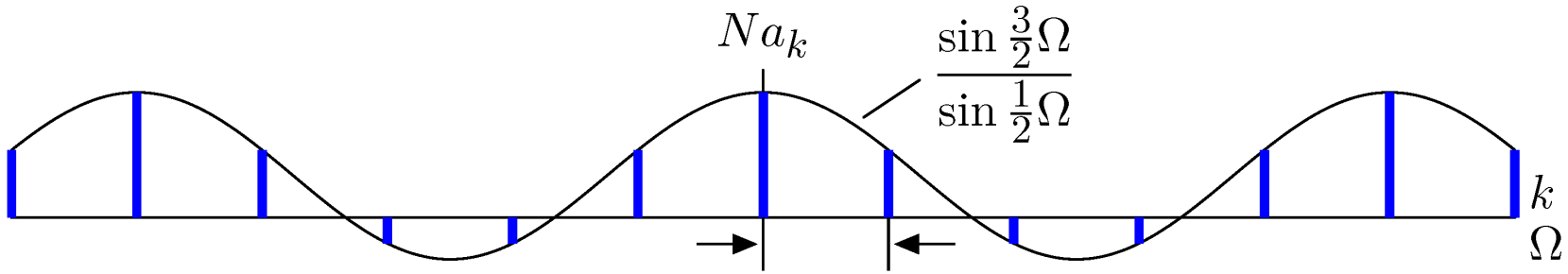
Then $x[n] = \lim_{N\to\infty} x_N[n]$.

Fourier Transform

Represent $x_N[n]$ by its Fourier series.



$$a_k = \frac{1}{N}\sum_{N} x_N[n]e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}\sum_{n=-N_1}^{N_1} e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}\,\frac{\sin\left(N_1+\frac{1}{2}\right)\Omega}{\sin\frac{1}{2}\Omega}$$

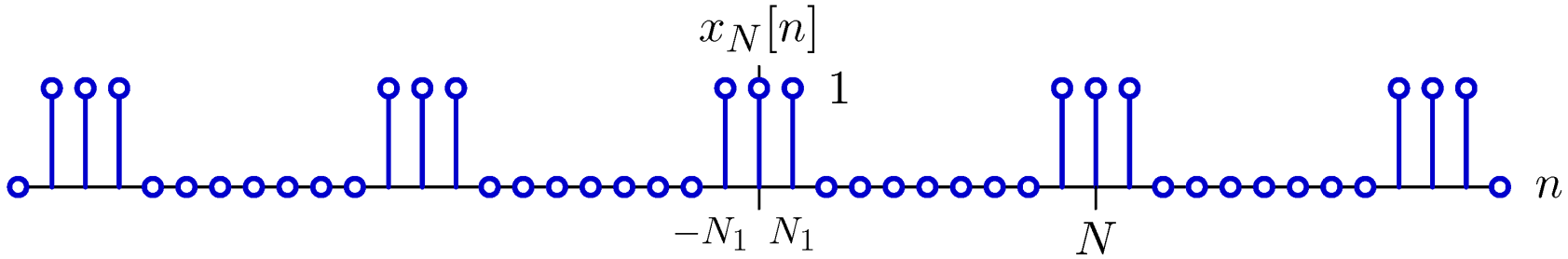


$$\Omega_0 = \frac{2\pi}{N}$$

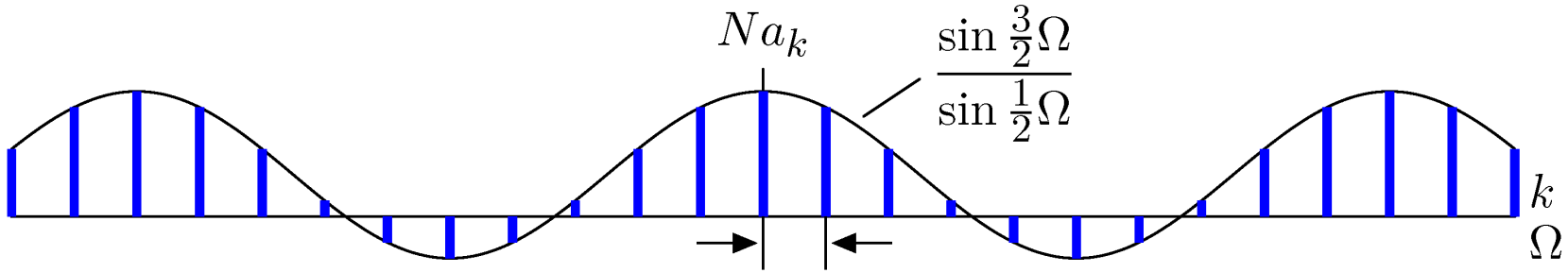
$$\Omega = k\Omega_0 = k\frac{2\pi}{N}$$

Fourier Transform

Doubling period doubles # of harmonics in given frequency interval.



$$a_k = \frac{1}{N}\sum_{N} x_N[n]e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}\sum_{n=-N_1}^{N_1} e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}\,\frac{\sin\left(N_1+\frac{1}{2}\right)\Omega}{\sin\frac{1}{2}\Omega}$$

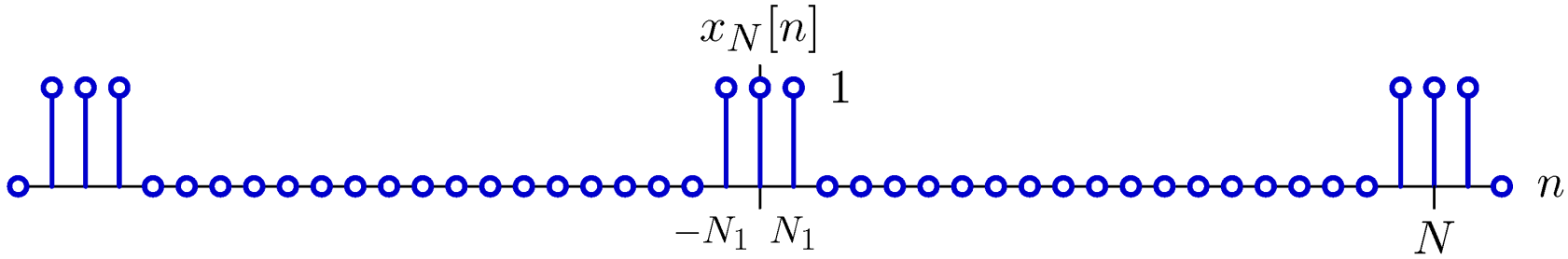


$$\Omega_0 = \frac{2\pi}{N}$$

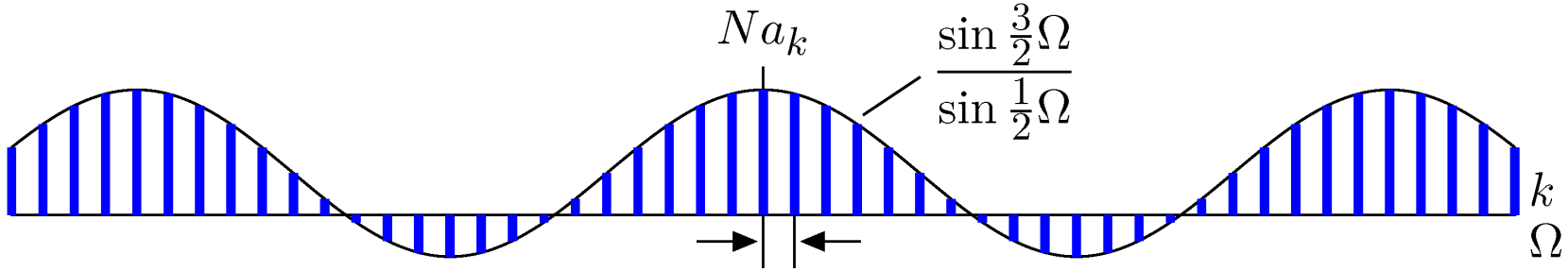
$$\Omega = k\Omega_0 = k\frac{2\pi}{N}$$

Fourier Transform

As $N \to \infty$, discrete harmonic amplitudes $\to$ a continuum $E(\Omega)$.



$$a_k = \frac{1}{N}\sum_{N} x_N[n]e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}\sum_{n=-N_1}^{N_1} e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}\,\frac{\sin\left(N_1+\frac{1}{2}\right)\Omega}{\sin\frac{1}{2}\Omega}$$

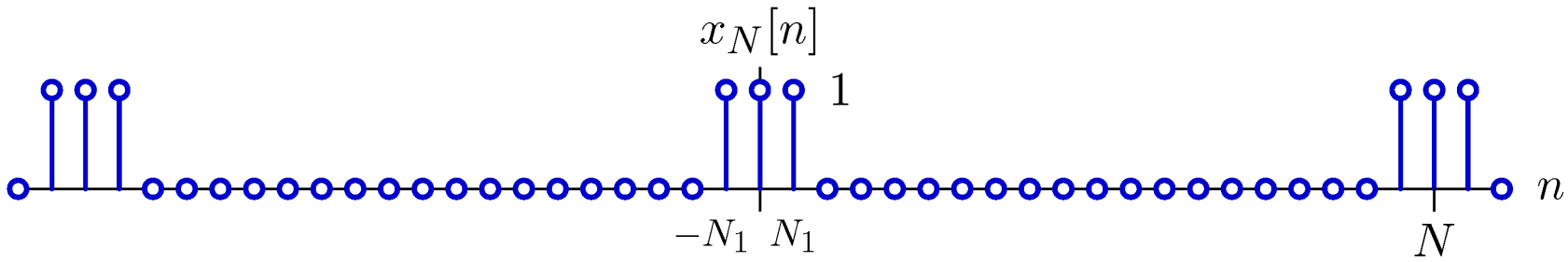


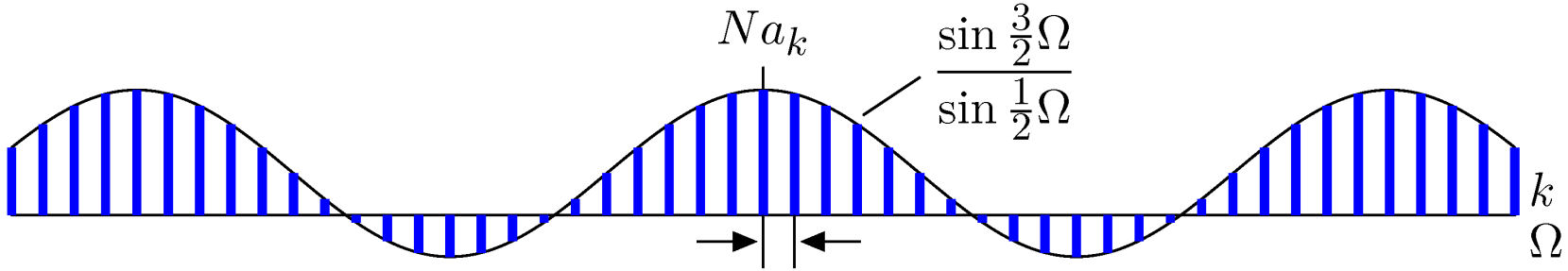
$$\Omega_0 = \frac{2\pi}{N} \qquad \Omega = k\Omega_0 = k\frac{2\pi}{N}$$

$$Na_k = \sum_{n=<N>} x[n]e^{-j\frac{2\pi}{N}kn} = \sum_{n=<N>} x[n]e^{-j\Omega n} = E(\Omega)$$

Fourier Transform

As $N \to \infty$, synthesis sum $\to$ integral.





$$\Omega_0 = \frac{2\pi}{N} \qquad \Omega = k\Omega_0 = k\frac{2\pi}{N}$$

$$Na_k = \sum_{n=<N>} x[n]e^{-j\frac{2\pi}{N}kn} = \sum_{n=<N>} x[n]e^{-j\Omega n} = E(\Omega)$$

$$x[n] = \sum_{k=<N>} \underbrace{\frac{1}{N}E(\Omega)}_{a_k} e^{j\frac{2\pi}{N}kn} = \sum_{k=<N>} \frac{\Omega_0}{2\pi}E(\Omega)e^{j\Omega n} \to \frac{1}{2\pi}\int_{2\pi} E(\Omega)e^{j\Omega n}d\Omega$$

Fourier Transform

Replacing $E(\Omega)$ by $X(e^{j\Omega})$ yields the DT Fourier transform relations.

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \qquad \text{("analysis" equation)}$$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\Omega})e^{j\Omega n}d\Omega \qquad \text{("synthesis" equation)}$$

Note for DTFT the *FT* and inverse *FT* are not symmetric.
One is integration over a finite interval (2π), and the
other is summation over infinite terms.

Relation between Fourier and Z Transforms

If the Z transform of a signal exists and if the ROC includes the unit circle, then the Fourier transform is equal to the Z transform evaluated on the unit circle.

Z transform:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

DT Fourier transform:

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} = H(z)\big|_{z=e^{j\Omega}}$$

Relation between Fourier and Z Transforms

Fourier transform "inherits" properties of Z transform.

Property	$x[n]$	$X(z)$	$X(e^{j\Omega})$
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(s) + bX_2(s)$	$aX_1(e^{j\Omega}) + bX_2(e^{j\Omega}$
Time shift	$x[n - n_0]$	$z^{-n_0}X(z)$	$e^{-j\Omega n_0}X(e^{j\Omega})$
Multiply by n	$nx[n]$	$-z\frac{d}{dz}X(z)$	$j\frac{d}{d\Omega}X(e^{j\Omega})$
Convolution	$(x_1 * x_2)[n]$	$X_1(z) \times X_2(z)$	$X_1(e^{j\Omega}) \times X_2(e^{j\Omega}$

Check Yourself

Ex.

DT Averager/Smoother

$$y[n] = \frac{1}{3}\{x[n-1] + x[n] + x[n+1]\}$$

Determine if it is a LPF or HPF?

Check Yourself

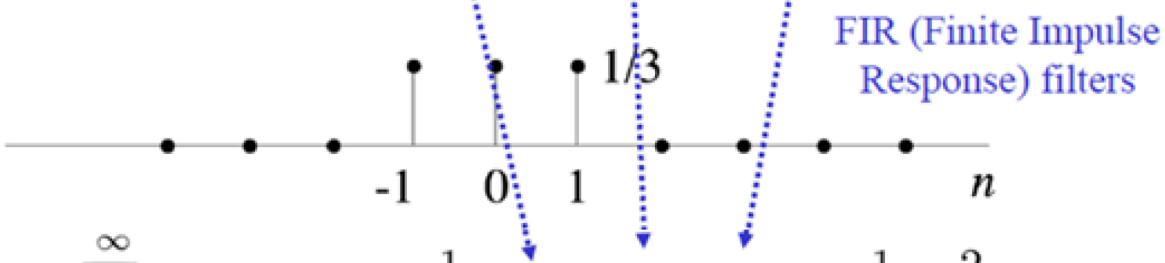
Ex.

DT Averager/Smoother

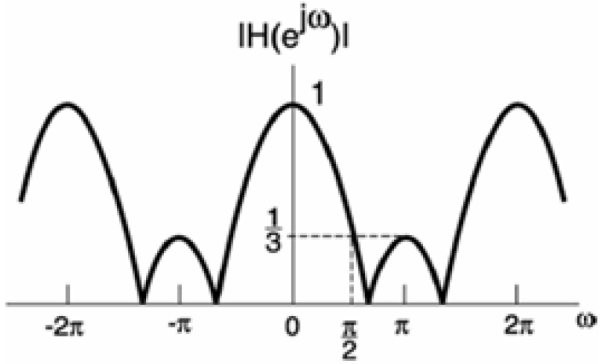
$$y[n] = \frac{1}{3}\{x[n-1] + x[n] + x[n+1]\}$$

Determine if it is a LPF or HPF?

$$h[n] = \frac{1}{3}\{\delta[n-1] + \delta[n] + \delta[n+1]\}$$



$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \frac{1}{3}\left[e^{-j\omega} + 1 + e^{j\omega}\right] = \frac{1}{3} + \frac{2}{3}\cos\omega$$



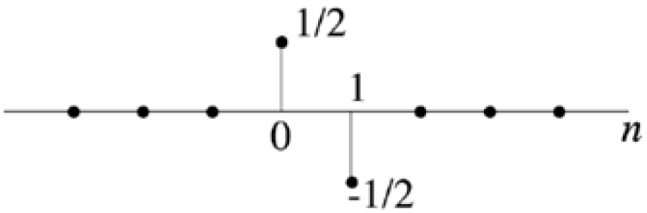
LPF

Check Yourself

Ex.

Simple DT "Edge" Detector
— DT 2-point "differentiator"

$$y[n] = \frac{1}{2}\{x[n] - x[n-1]\}$$



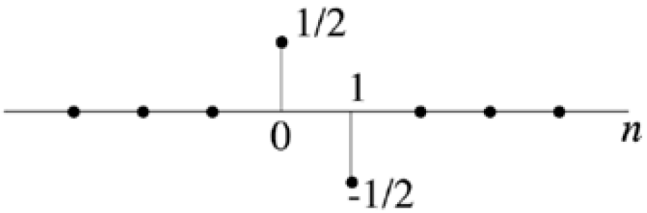
Determine if it is a LPF or HPF?

Check Yourself

Ex.

Simple DT "Edge" Detector
— DT 2-point "differentiator"

$$y[n] = \frac{1}{2}\{x[n] - x[n-1]\}$$



Determine if it is a LPF or HPF?

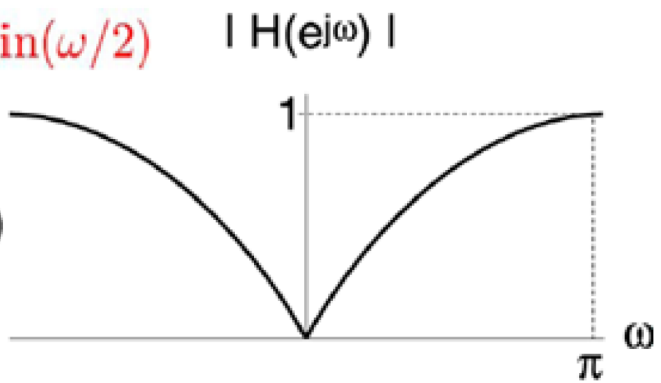
$$h[n] = \frac{1}{2}\{\delta[n] - \delta[n-1]\}$$

$$\frac{j}{2j}e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2}) = je^{-j\omega/2}\sin(\omega/2)$$

$$\Downarrow$$

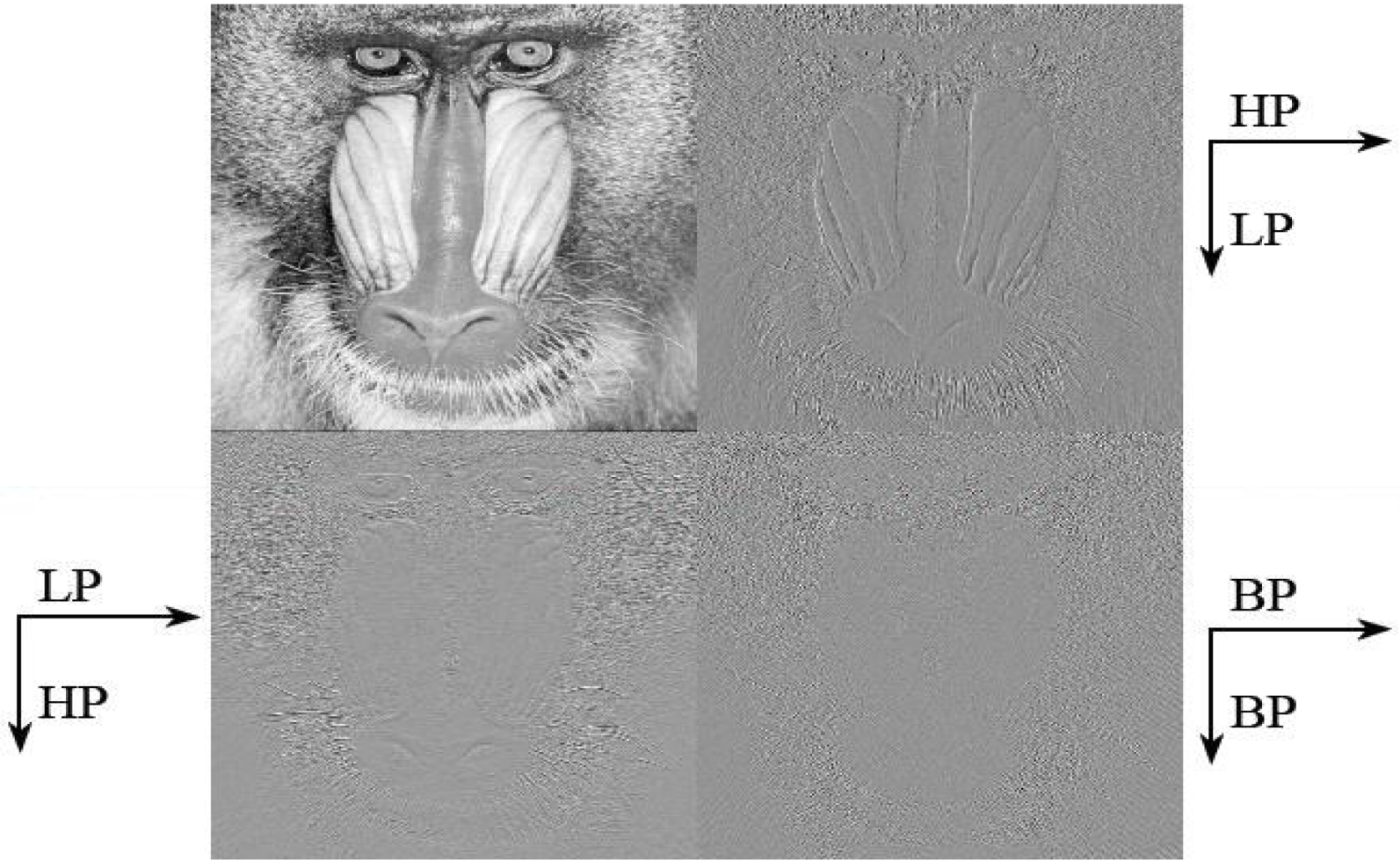
$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega}) = je^{-j\omega/2}\sin(\omega/2)$$

$$|H(e^{j\omega})| = |\sin(\omega/2)|$$



DT Fourier Transform of Images

Demo: Apply different filters to two-dimensional image signals.

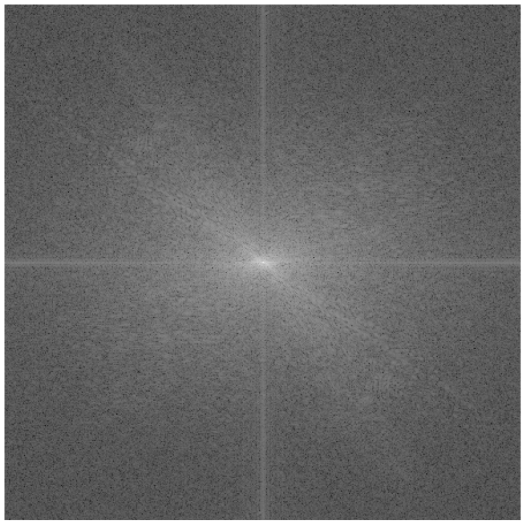
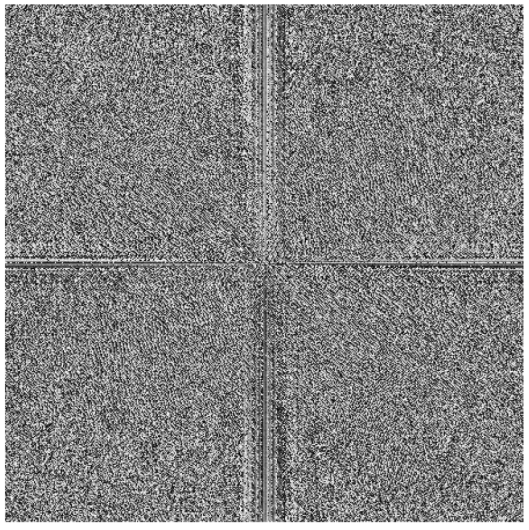
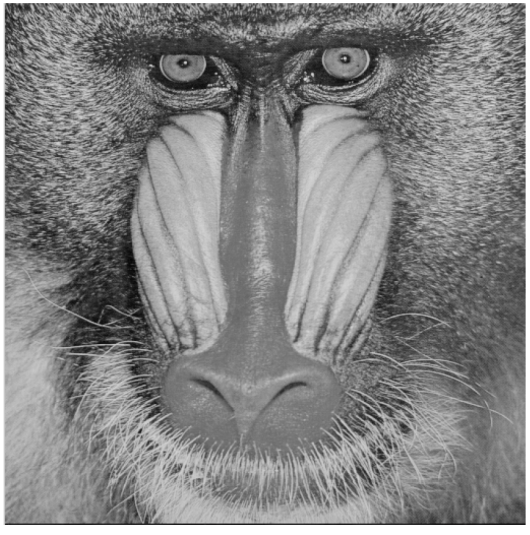
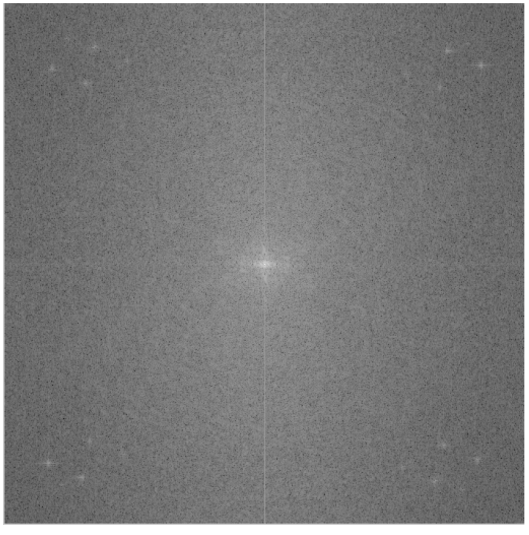
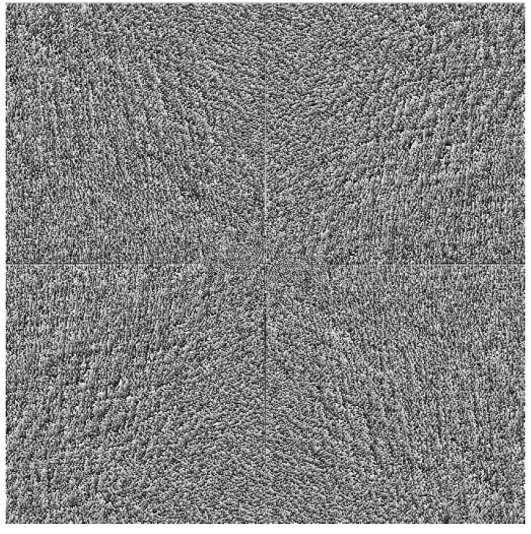


DT Fourier Transform of Images

Demo: Apply different filters to two-dimensional image signals.



DT Fourier Transform of Images

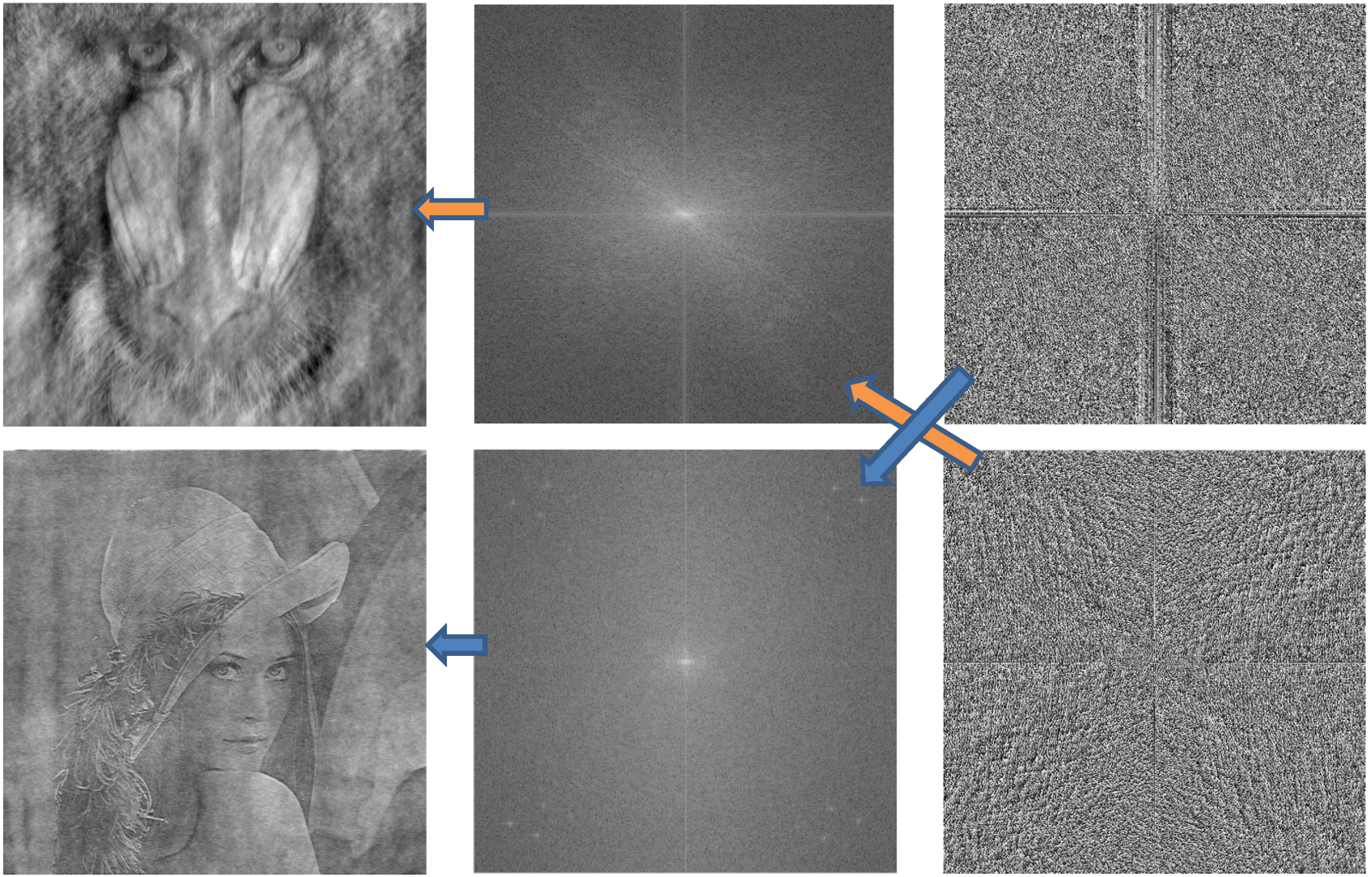







Image

Log of magnitude

Phase

DT Fourier Transform of Images



Four Fourier Representations

- CT Fourier series
- CT Fourier transform
- DT Fourier series
- DT Fourier transform

Relations among the four Fourier representations.

Four Fourier Representations

We have discussed four closely related Fourier representations.

DT Fourier Series

$$a_k = a_{k+N} = \frac{1}{N} \sum_{n=<N>} x[n] e^{-j\frac{2\pi}{N}kn}$$

$$x[n] = x[n+N] = \sum_{k=<N>} a_k e^{j\frac{2\pi}{N}kn}$$

DT Fourier transform

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{<2\pi>} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$

CT Fourier Series

$$a_k = \frac{1}{T} \int_T x(t) e^{-j\frac{2\pi}{T}kt} dt$$

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt}$$

CT Fourier transform

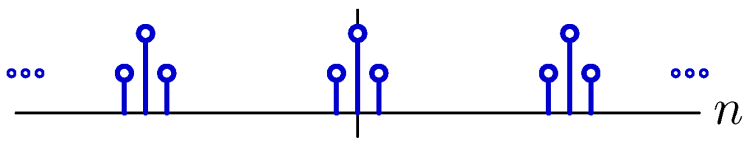
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

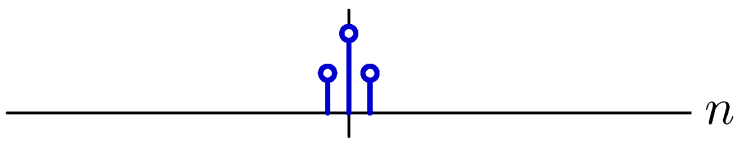
Four Types of Time

discrete vs. continuous ($\updownarrow$) and periodic vs aperiodic ($\leftrightarrow$)

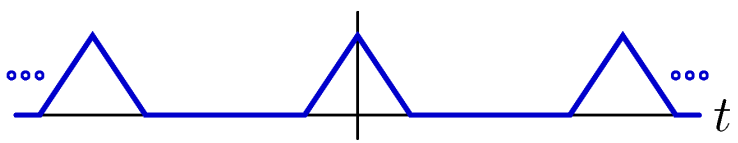
DT Fourier Series



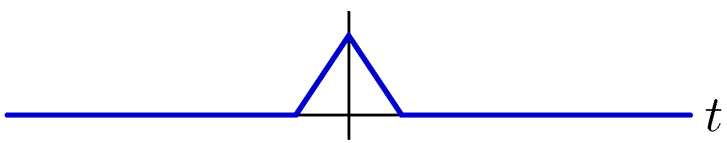
DT Fourier transform



CT Fourier Series



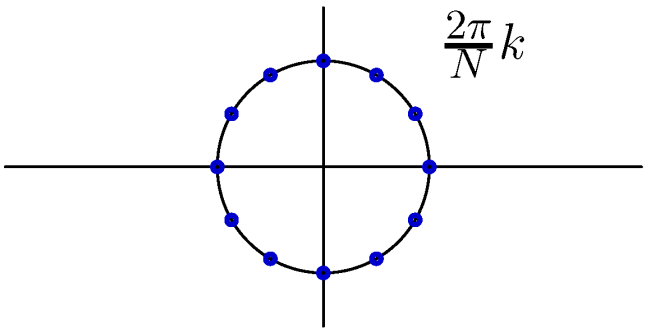
CT Fourier transform



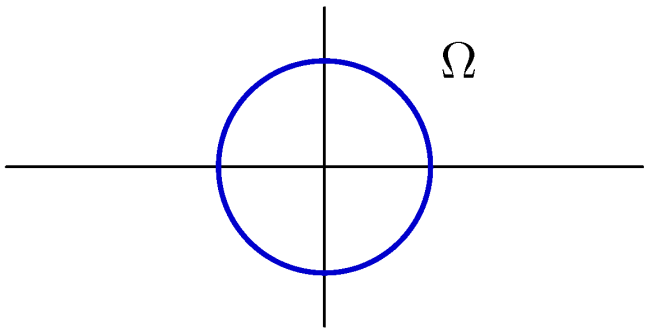
Four Types of Frequency

discrete vs. continuous ($\leftrightarrow$) and periodic vs aperiodic ($\updownarrow$)

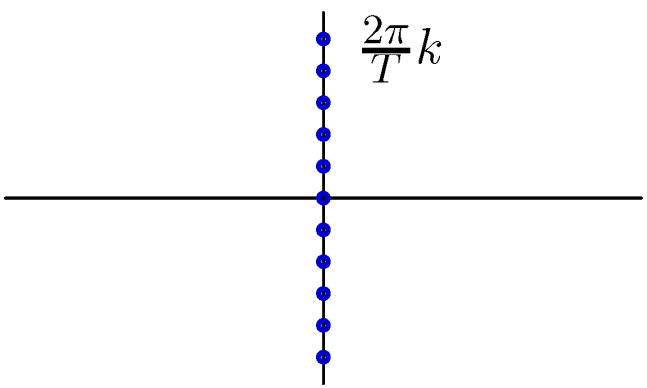
DT Fourier Series



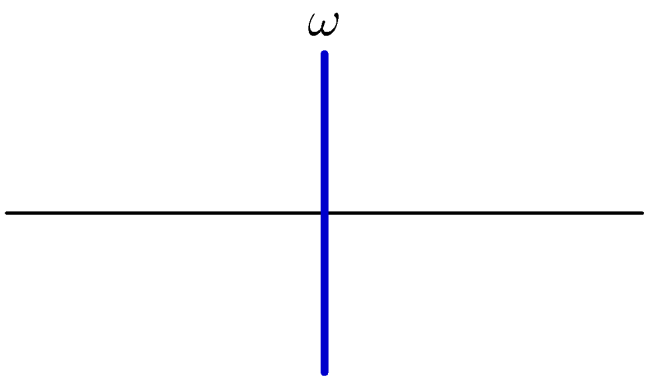
DT Fourier transform



CT Fourier Series



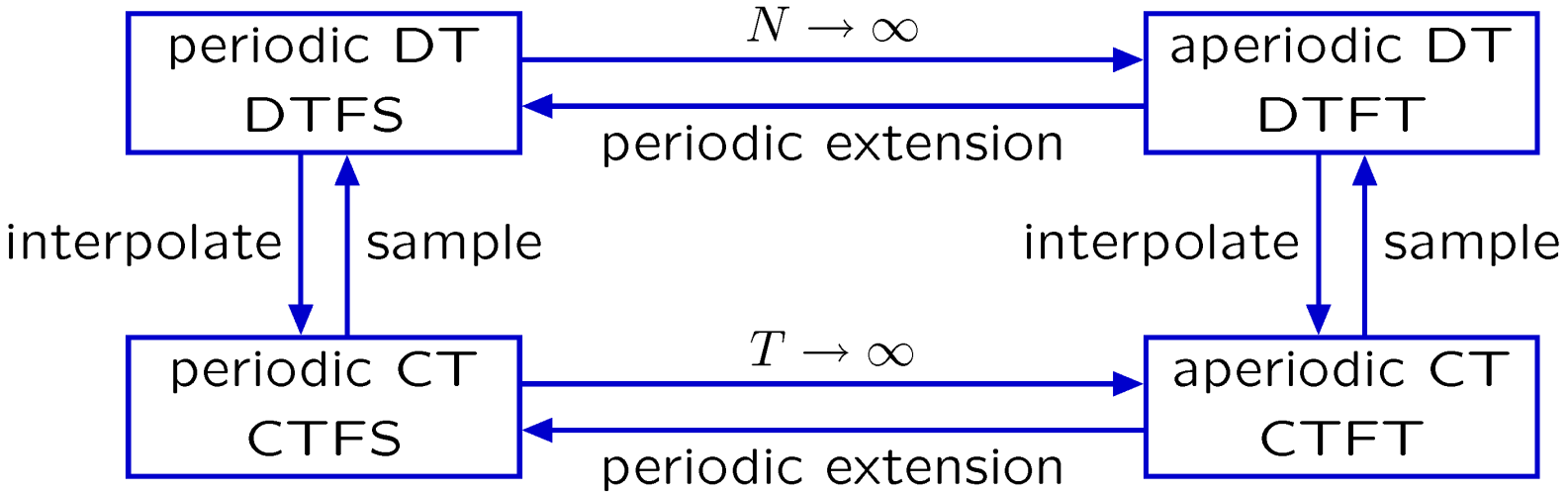
CT Fourier transform



Relations among Fourier Representations

Different Fourier representations are related because they apply to signals that are related.

DTFS (discrete-time Fourier series):	periodic DT
DTFT (discrete-time Fourier transform):	aperiodic DT
CTFS (continuous-time Fourier series):	periodic CT
CTFT (continuous-time Fourier transform):	aperiodic CT



Relation between Fourier Series and Transform

A periodic signal can be represented by a Fourier series or by an equivalent Fourier transform.

Series: represent periodic signal as weighted sum of harmonics

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 kt} \; ; \qquad \omega_0 = \frac{2\pi}{T}$$

The Fourier transform of a sum is the sum of the Fourier transforms:

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

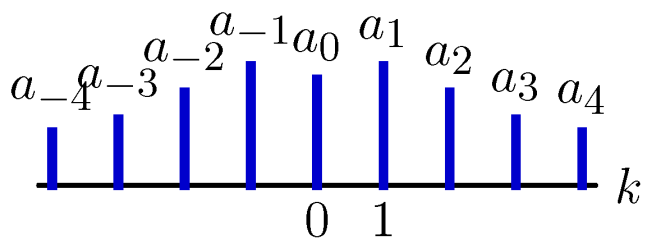
Therefore periodic signals can be equivalently represented as Fourier transforms (with impulses!).

Relation between Fourier Series and Transform

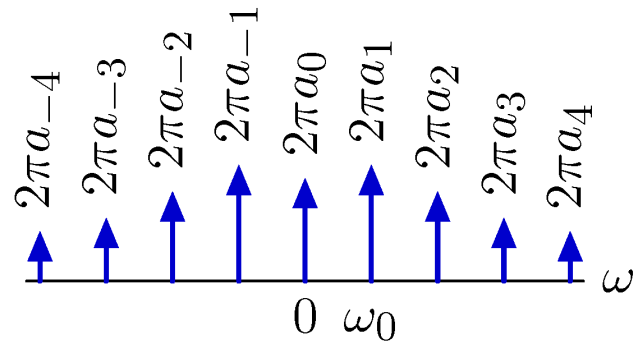
A periodic signal can be represented by a Fourier series or by an equivalent Fourier transform.

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 kt} \quad \leftrightarrow$$

Fourier Series



Fourier Transform

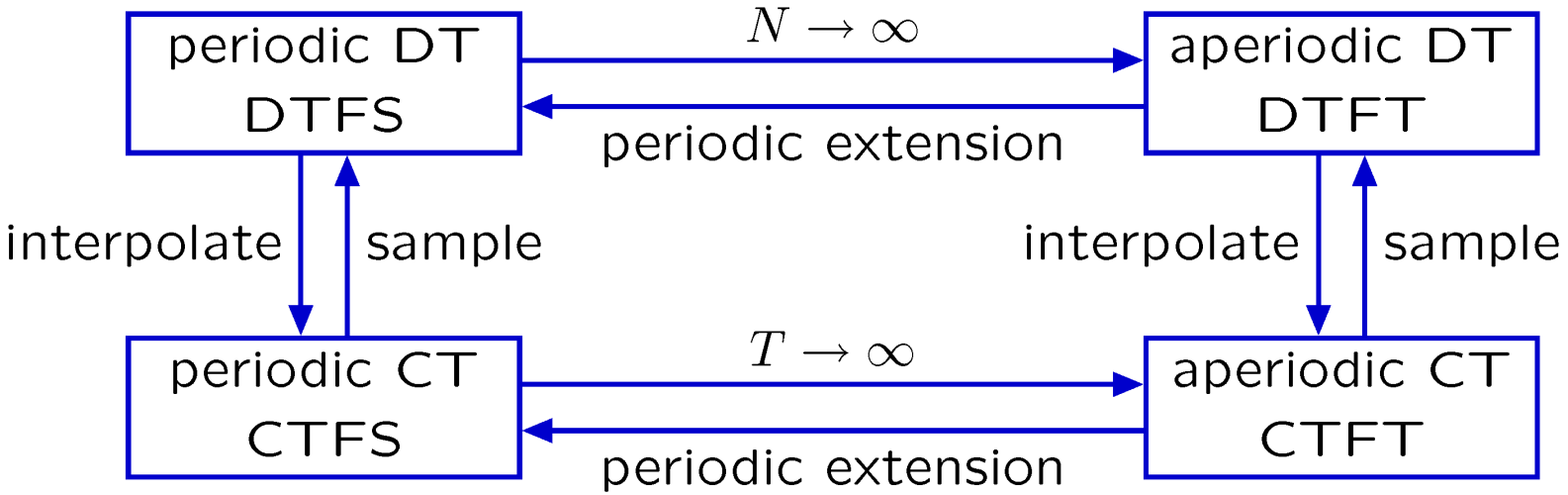


Relations among Fourier Representations

Explore other relations among Fourier representations.

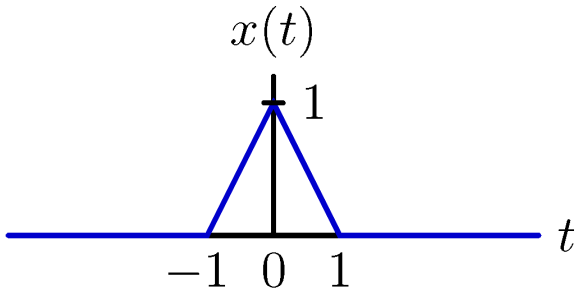
Start with an aperiodic CT signal. Determine its Fourier transform.

Convert the signal so that it can be represented by alternate Fourier representations and compare.



Start with the CT Fourier Transform

Determine the Fourier transform of the following signal.

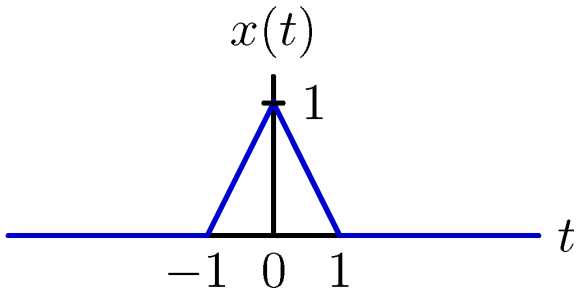


Could calculate Fourier transform from the definition.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{j\omega t}dt$$

Start with the CT Fourier Transform

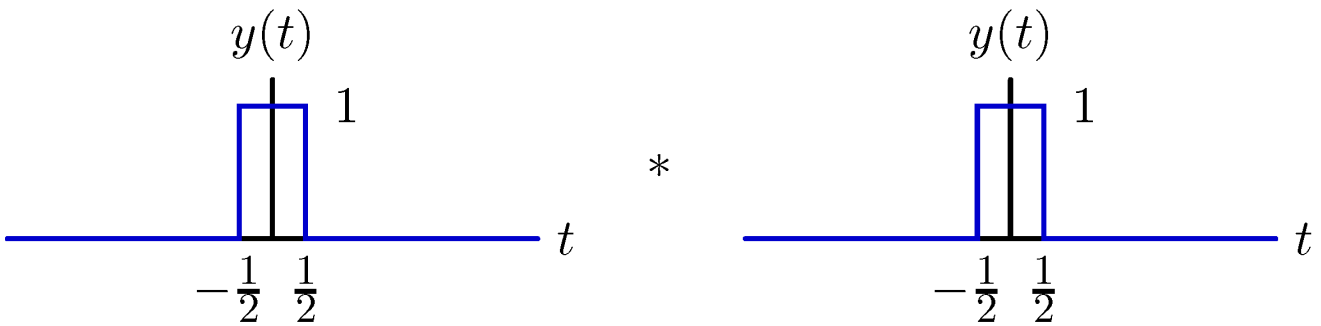
Determine the Fourier transform of the following signal.



Could calculate Fourier transform from the definition.

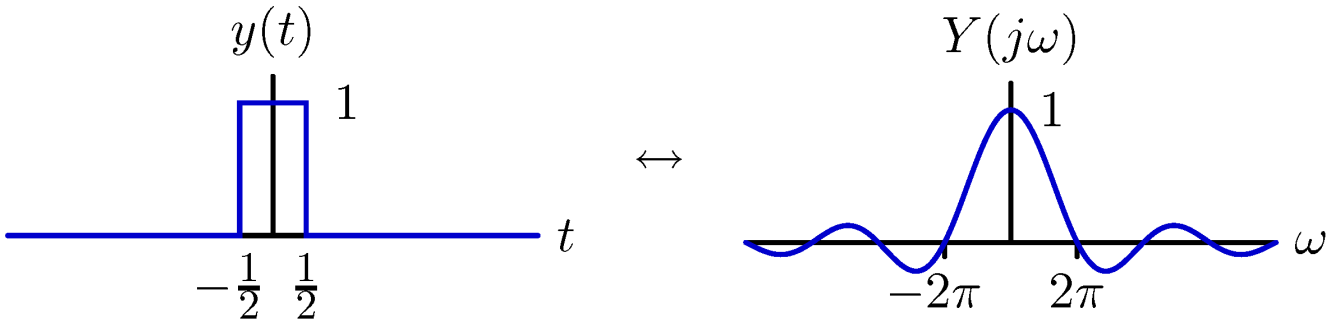
$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{j\omega t}dt$$

Easier to calculate $x(t)$ by convolution of two square pulses:

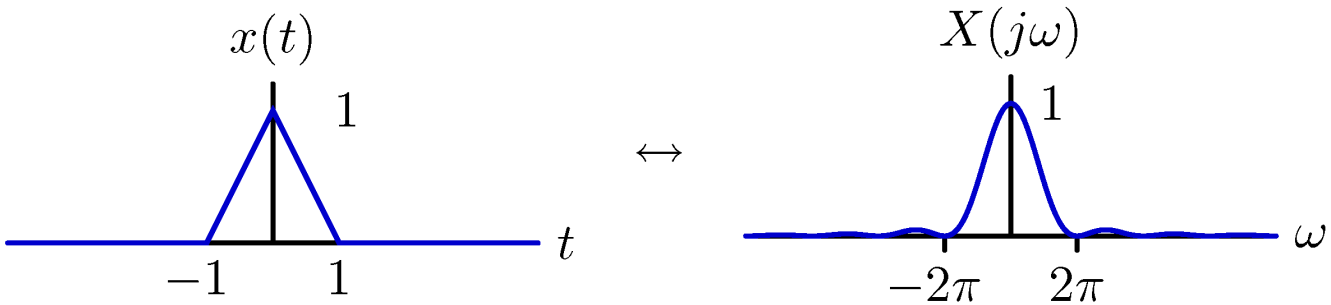


Start with the CT Fourier Transform

The transform of $y(t)$ is $\frac{2\sin(\omega/2)}{\omega}$



so the transform of $x(t) = (y * y)(t)$ is $X(j\omega) = Y(j\omega) \times Y(j\omega)$.

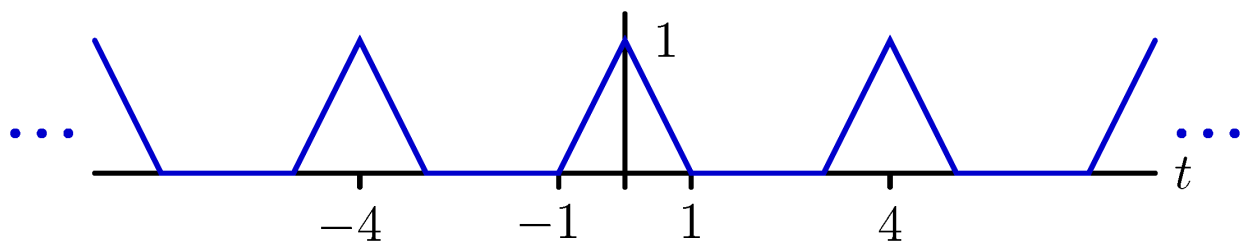


Relation between Fourier Transform and Series

What is the effect of making a signal periodic in time?

Find Fourier transform of periodic extension of $x(t)$ to period $T = 4$.

$$z(t) = \sum_{k=-\infty}^{\infty} x(t + 4k)$$

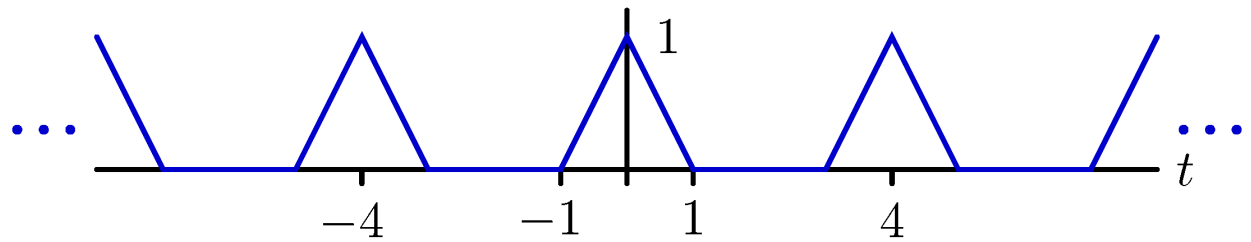


Could calculate $Z(j\omega)$ for the definition ... ugly.

Relation between Fourier Transform and Series

Easier to calculate $z(t)$ by convolving $x(t)$ with an impulse train.

$$z(t) = \sum_{k=-\infty}^{\infty} x(t+4k)$$



$$z(t) = \sum_{k=-\infty}^{\infty} x(t+4k) = (x * p)(t)$$

where

$$p(t) = \sum_{k=-\infty}^{\infty} \delta(t+4k)$$

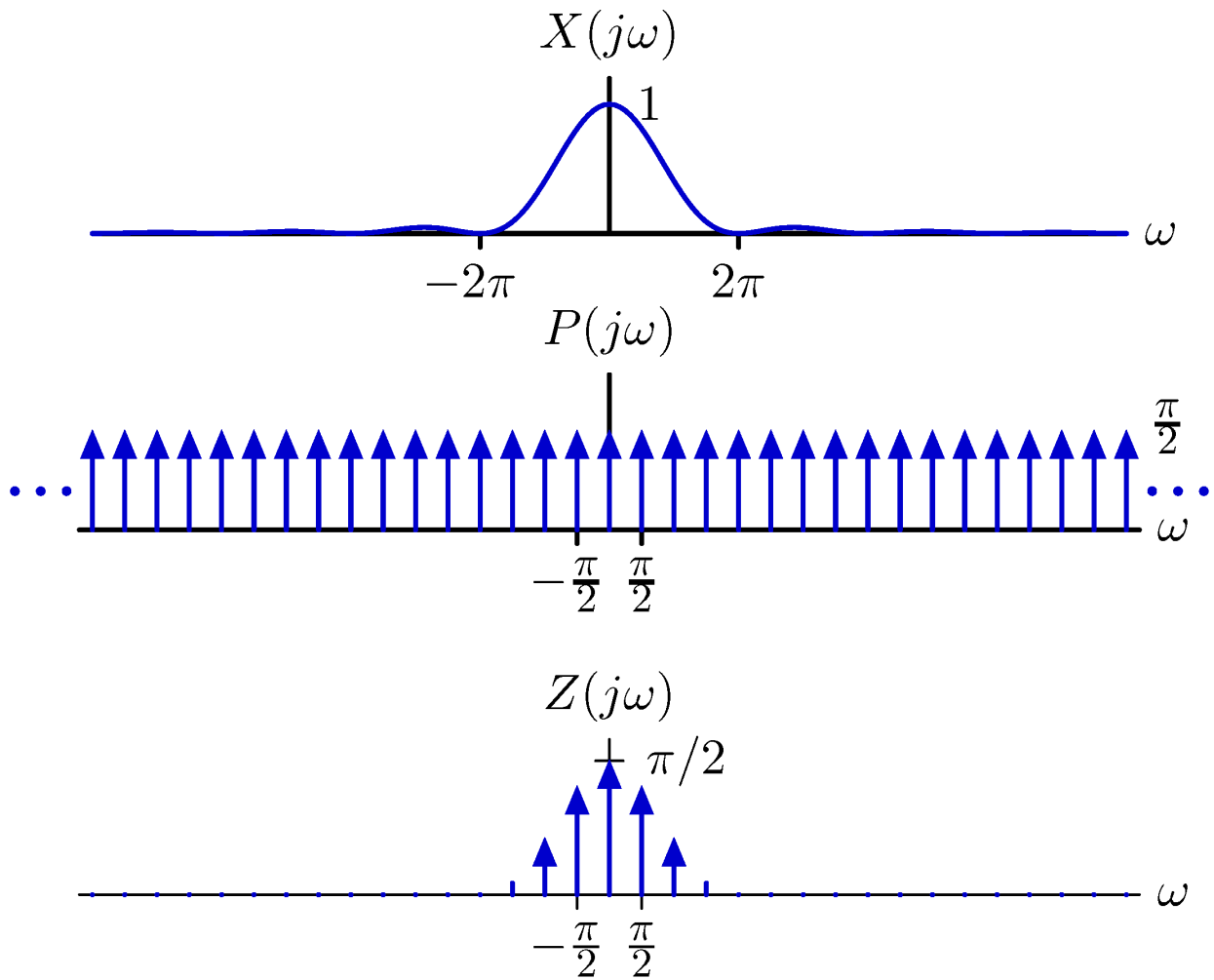
Then

$$Z(j\omega) = X(j\omega) \times P(j\omega)$$

We already know $P(j\omega)$: it's also an impulse train!

Relation between Fourier Transform and Series

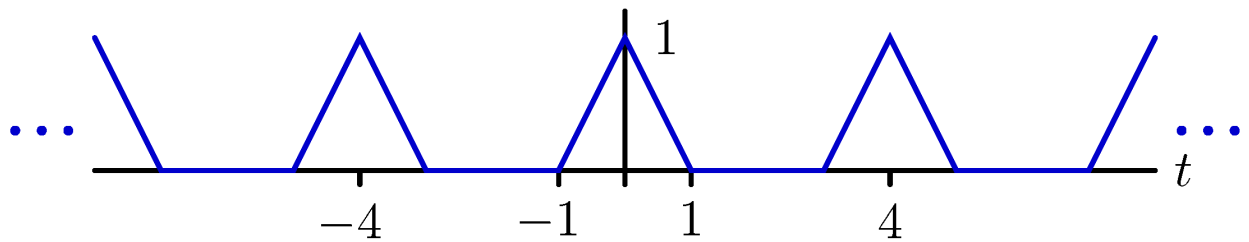
Convolving in time corresponds to multiplying in frequency.

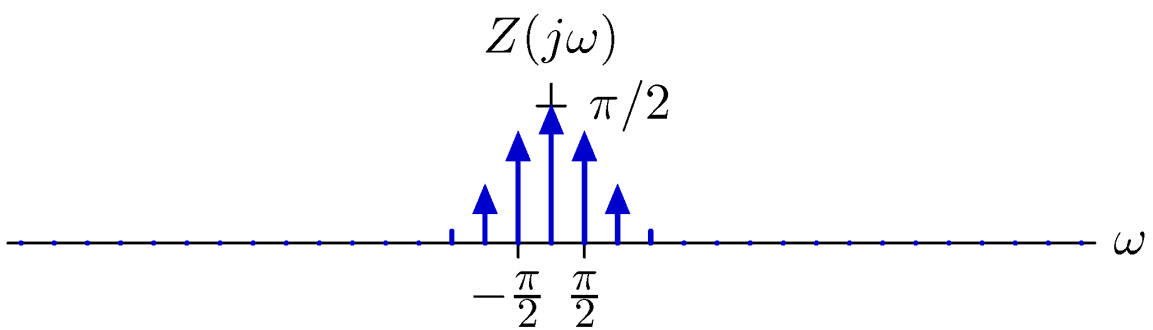


Relation between Fourier Transform and Series

The Fourier transform of a periodically extended function is a discrete function of frequency ω.

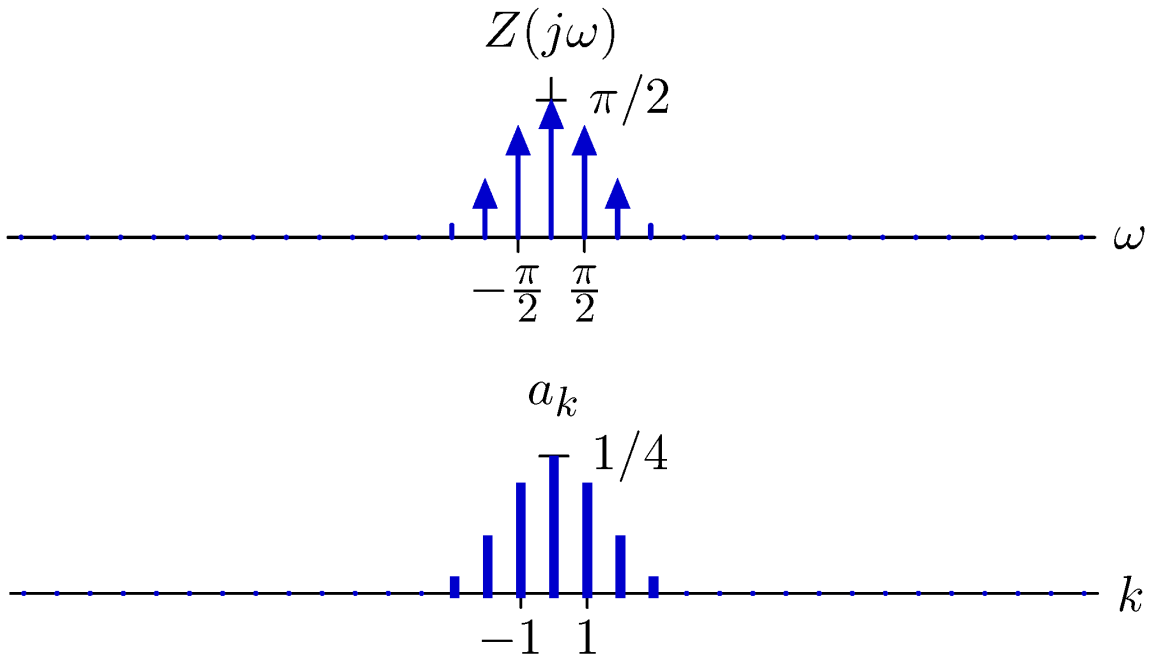
$$z(t) = \sum_{k=-\infty}^{\infty} x(t+4k)$$





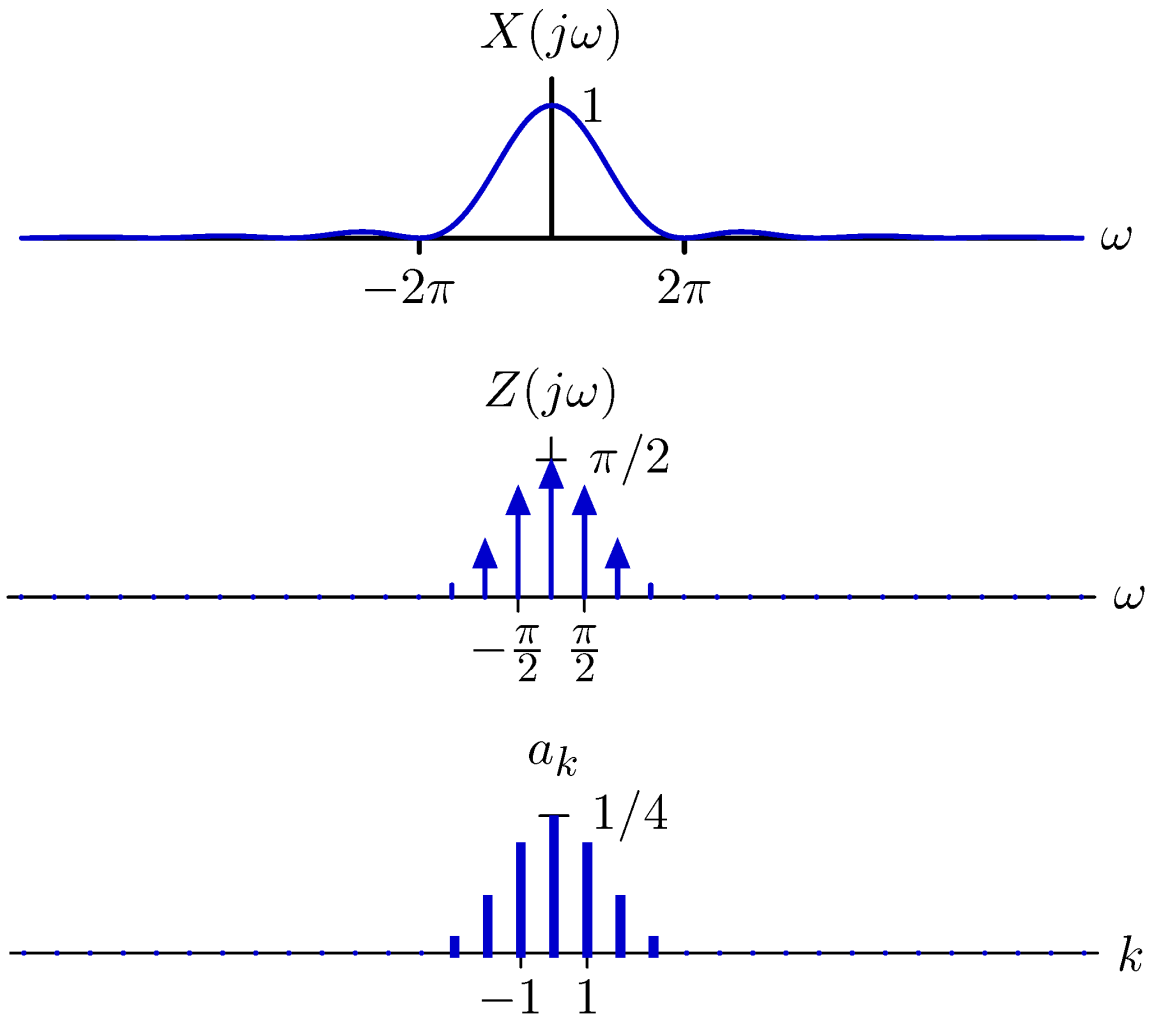
Relation between Fourier Transform and Series

The weight (area) of each impulse in the Fourier transform of a periodically extended function is 2π times the corresponding Fourier series coefficient.



Relation between Fourier Transform and Series

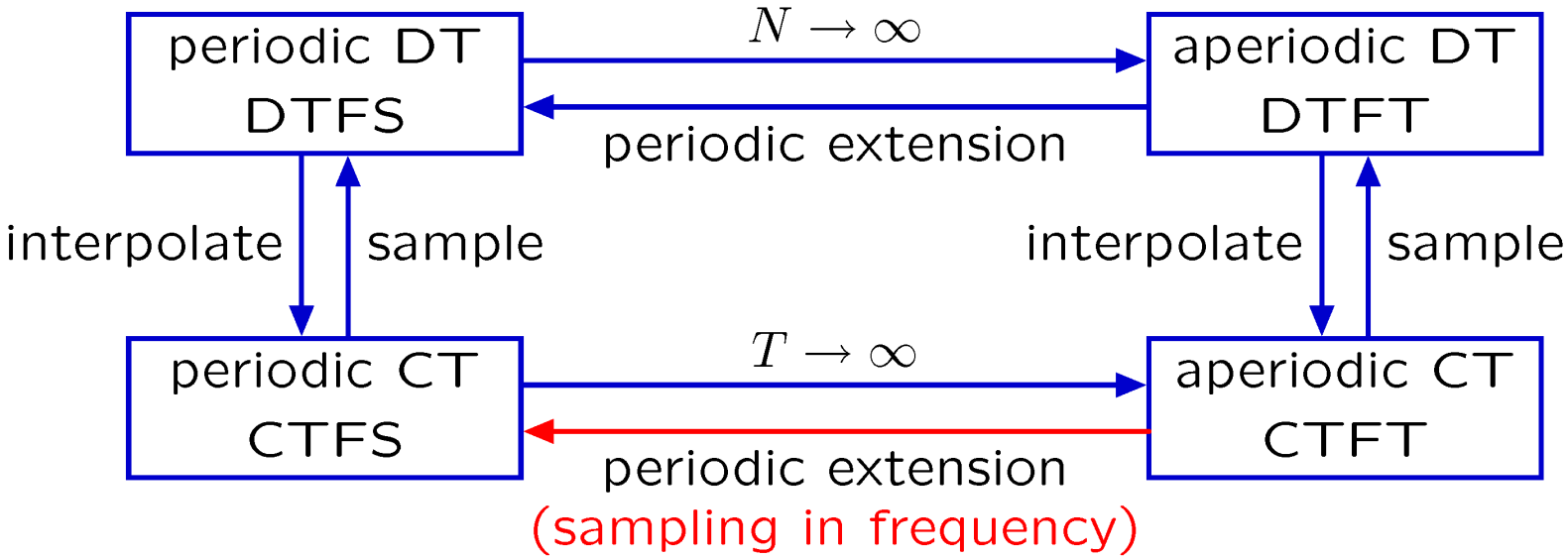
The effect of periodic extension of $x(t)$ to $z(t)$ is to sample the frequency representation.



Relation between Fourier Transform and Series

Periodic extension of a CT signal produces a discrete function of frequency.

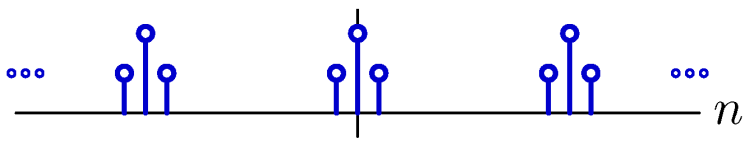
Periodic extension
= convolving with impulse train in time
= multiplying by impulse train in frequency
→ sampling in frequency



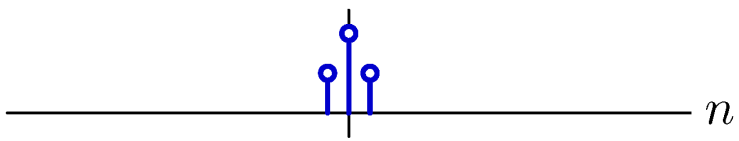
Four Types of Time

discrete vs. continuous ($\updownarrow$) and periodic vs aperiodic ($\leftrightarrow$)

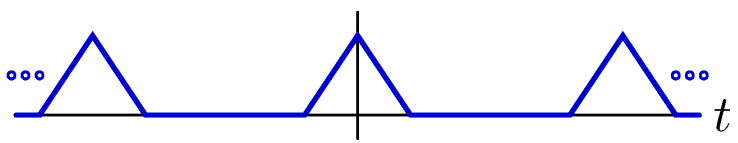
DT Fourier Series



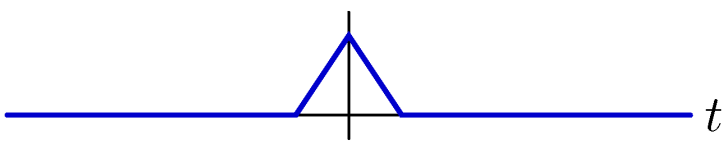
DT Fourier transform



CT Fourier Series



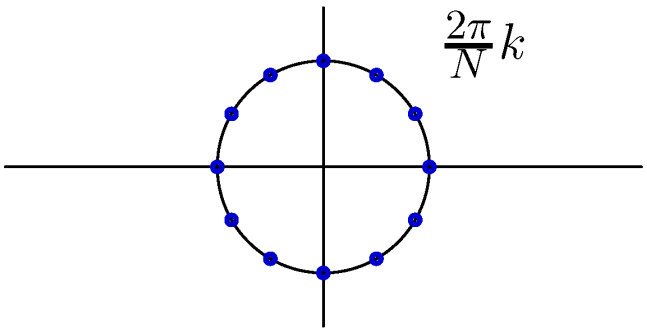
CT Fourier transform



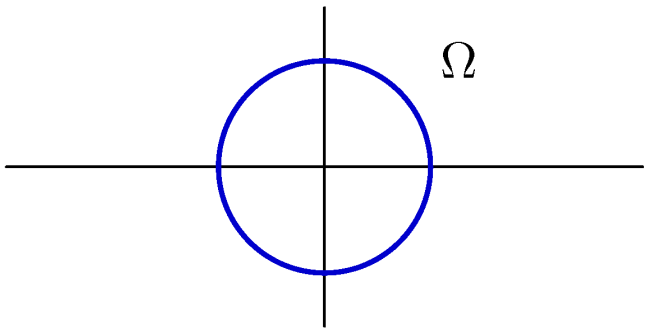
Four Types of Frequency

discrete vs. continuous ($\leftrightarrow$) and periodic vs aperiodic ($\updownarrow$)

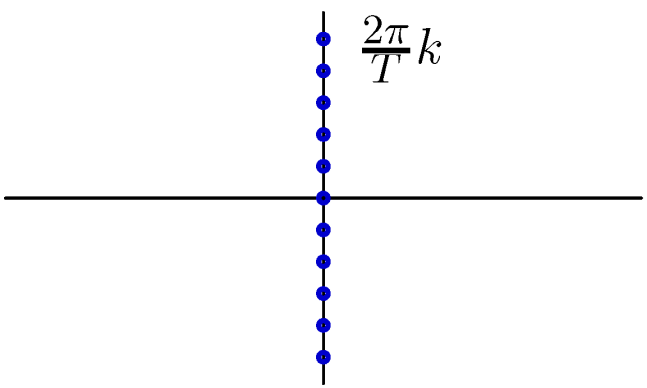
DT Fourier Series



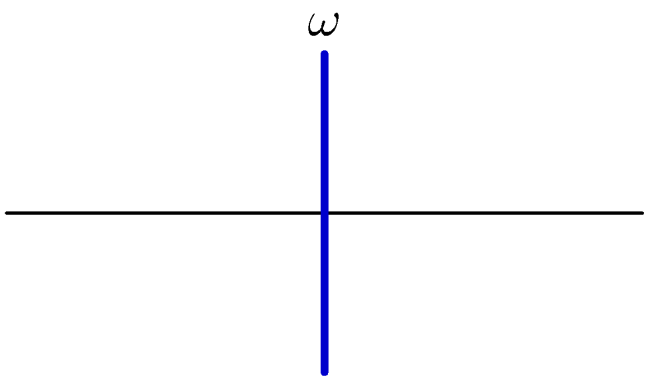
DT Fourier transform



CT Fourier Series

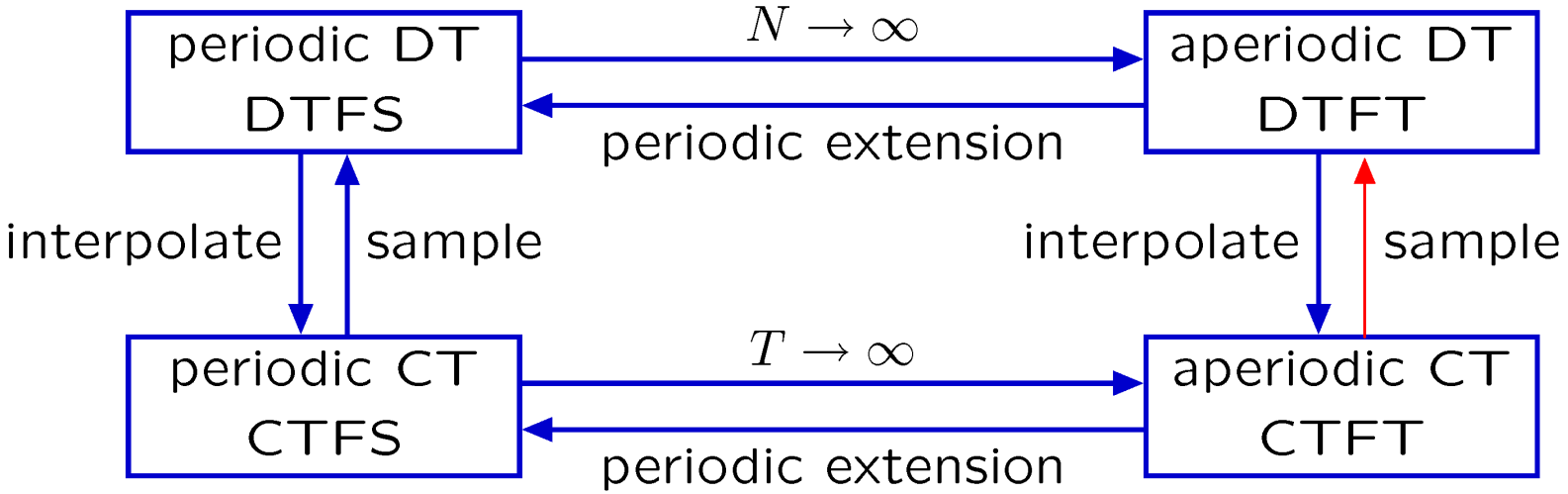


CT Fourier transform



Relations among Fourier Representations

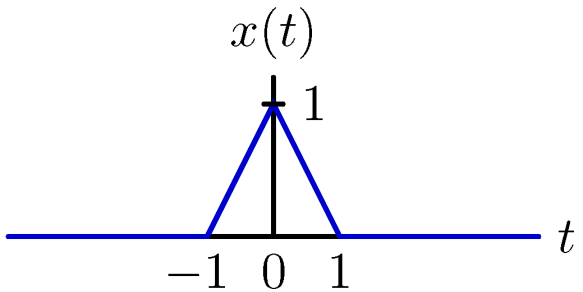
Compare to sampling in time.



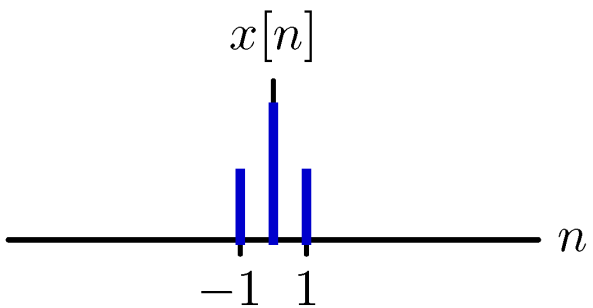
Relations between CT and DT transforms

Sampling a CT signal generates a DT signal.

$$x[n] = x(nT)$$



Take $T = \frac{1}{2}$.

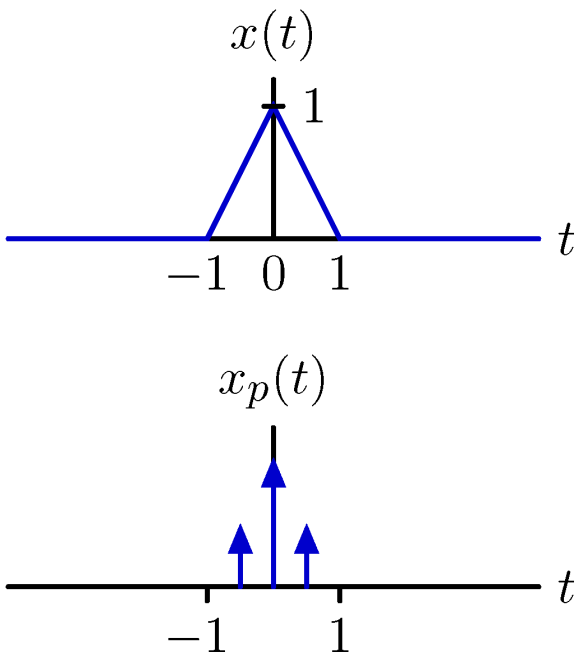


What is the effect on the frequency representation?

Relations between CT and DT transforms

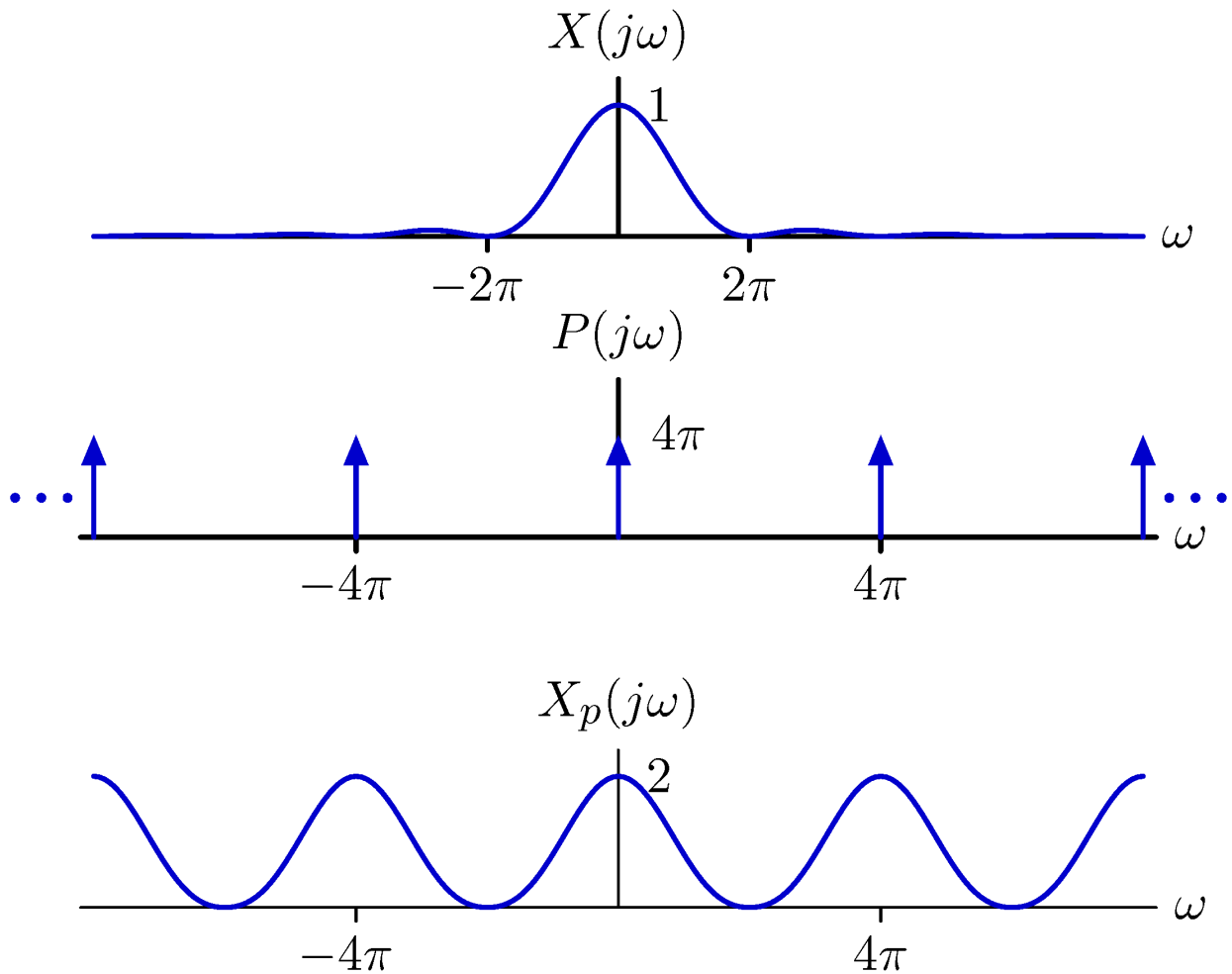
We can generate a signal with the same shape by multiplying $x(t)$ by an impulse train with $T = \frac{1}{2}$.

$$x_p(t) = x(t) \times p(t) \quad \text{where} \quad p(t) = \sum_{k=-\infty}^{\infty} \delta(t + kT)$$



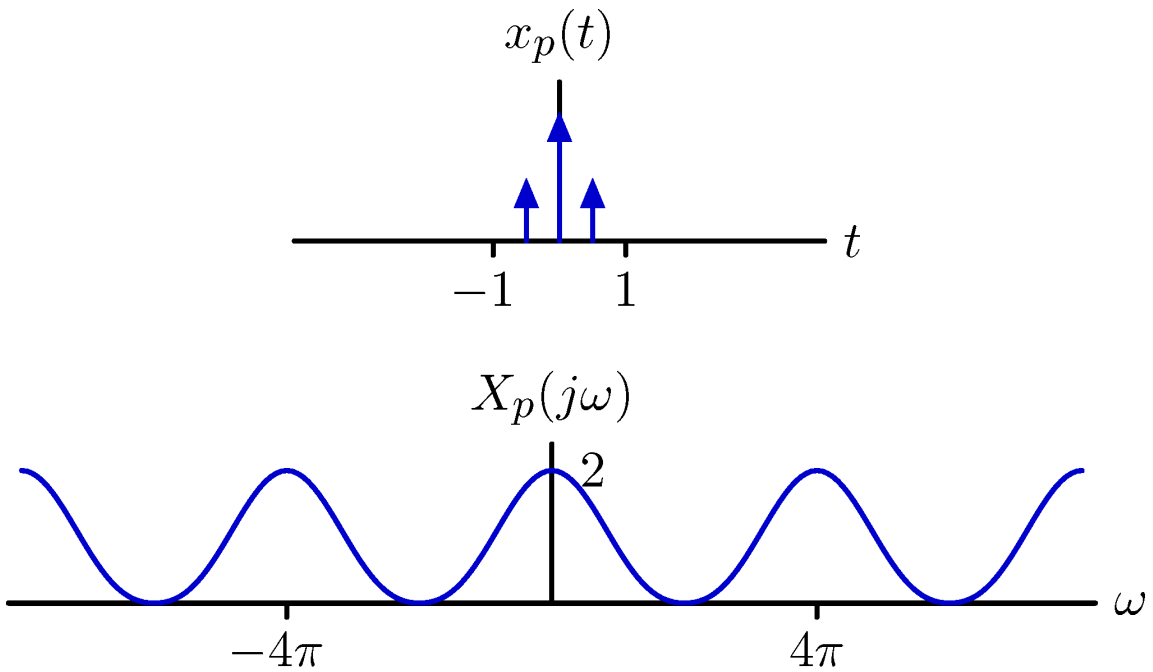
Relations between CT and DT transforms

Multiplying $x(t)$ by an impulse train in time is equivalent to convolving $X(j\omega)$ by an impulse train in frequency (then $\div 2\pi$).



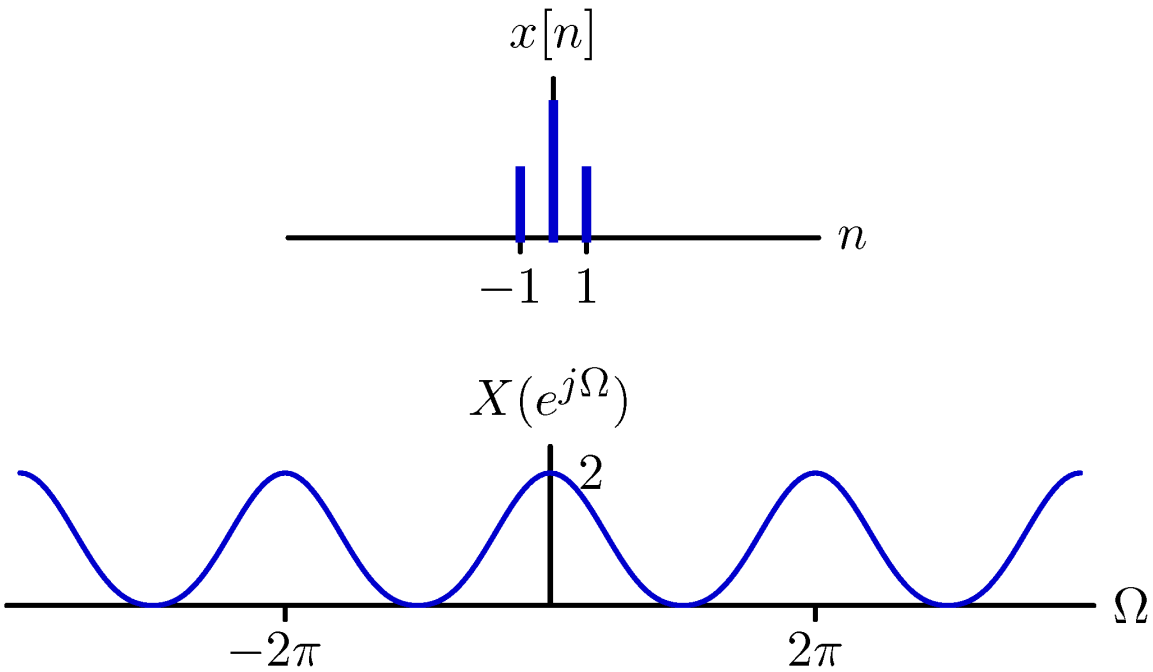
Relations between CT and DT transforms

The Fourier transform of the "sampled" signal $x_p(t)$ is periodic in ω with period 4π.



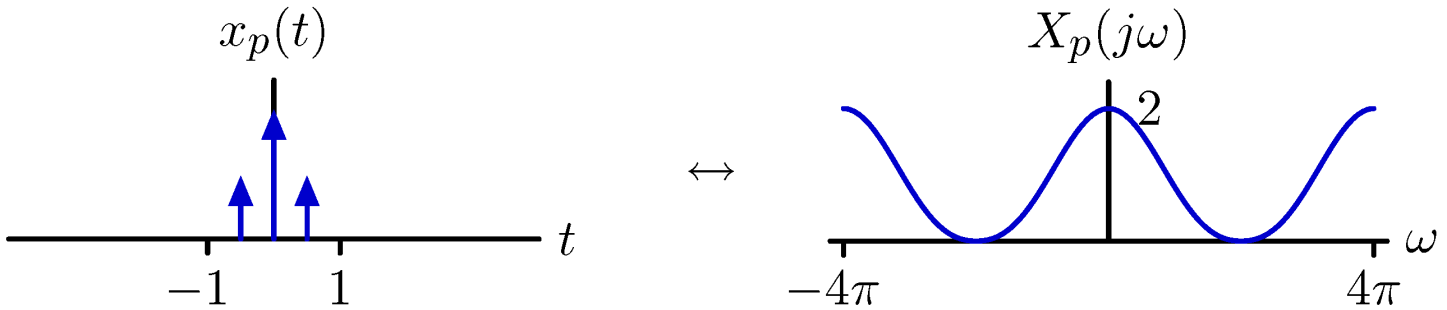
Relations between CT and DT transforms

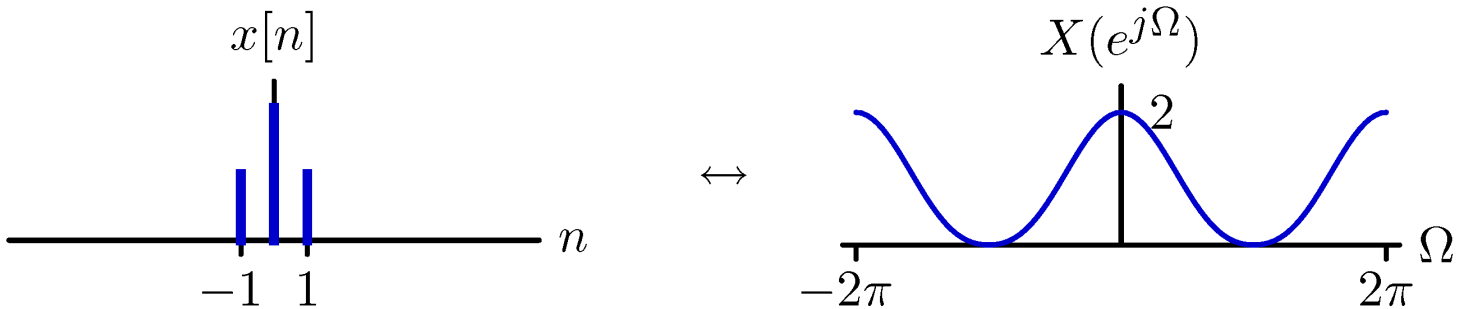
The Fourier transform of the "sampled" signal $x_p(t)$ has the same shape as the DT Fourier transform of $x[n]$.



DT Fourier transform

The CT Fourier transform of a "sampled" signal ($x_p(t)$) is equal to the DT Fourier transform of the samples ($x[n]$) where $\Omega = \omega T$, i.e., $X(j\omega) = X(e^{j\Omega})\Big|_{\Omega=\omega T}$.





$\Omega = \omega T = \frac{1}{2}\omega$

DT Fourier transform

Compare the definitions:

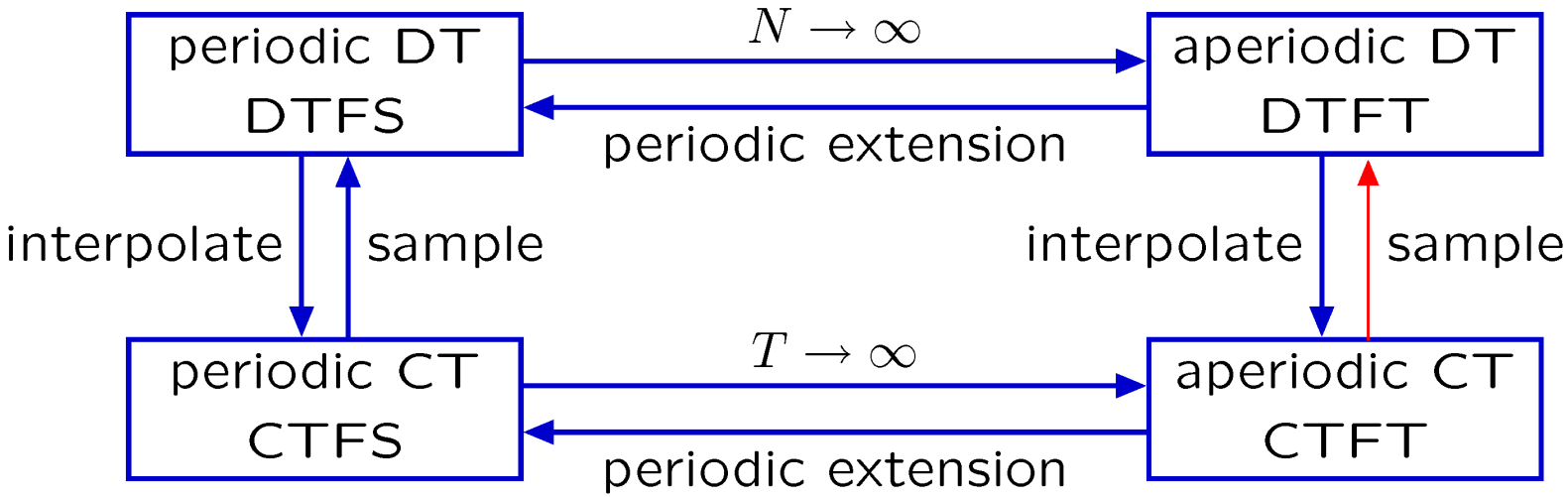
$$X(e^{j\Omega}) = \sum_n x[n]e^{-j\Omega n}$$

$$\begin{aligned} X_p(j\omega) &= \int x_p(t)e^{-j\omega t}dt \\ &= \int \sum_n x[n]\delta(t-nT)e^{-j\omega t}dt \\ &= \sum_n x[n] \int \delta(t-nT)e^{-j\omega t}dt \\ &= \sum_n x[n]e^{-j\omega nT} \end{aligned}$$

$$\Omega = \omega T$$

Relation between CT and DT Fourier transforms

The CT Fourier transform of a "sampled" signal ($x_p(t)$) is equal to the DT Fourier transform of the samples ($x[n]$) where $\Omega = \omega T$, i.e., $X(j\omega) = X(e^{j\Omega})\Big|_{\Omega=\omega T}$.



Relation Among Fourier Representations

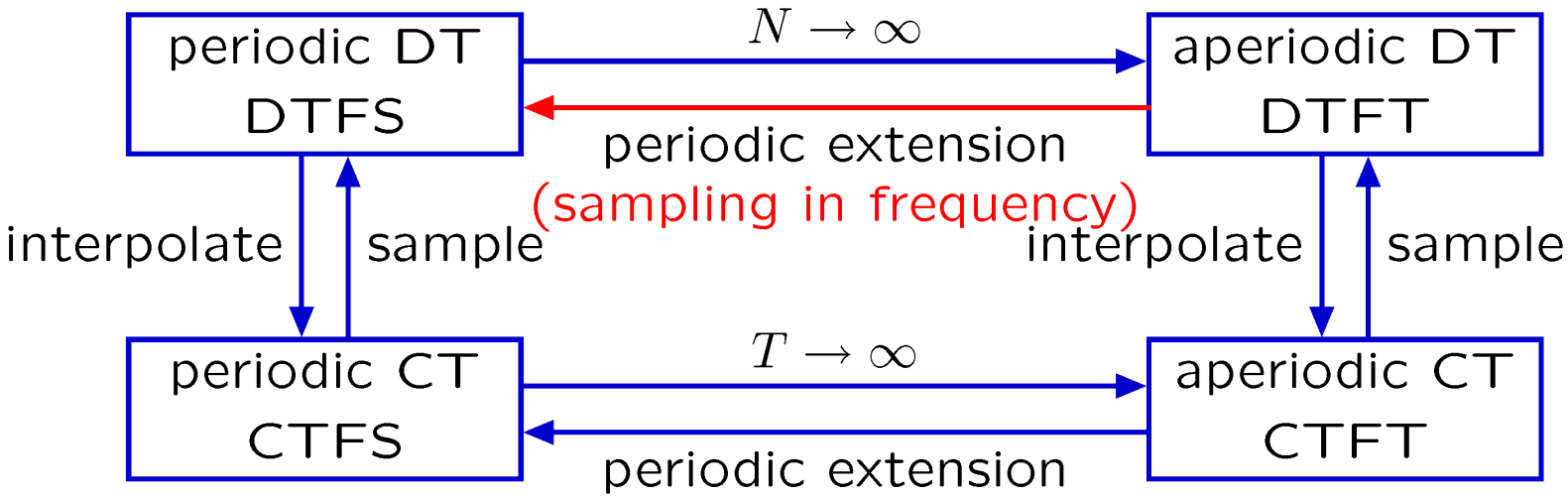
Periodic extension of a DT signal produces a discrete function of frequency.

Periodic extension

= convolving with impulse train in time

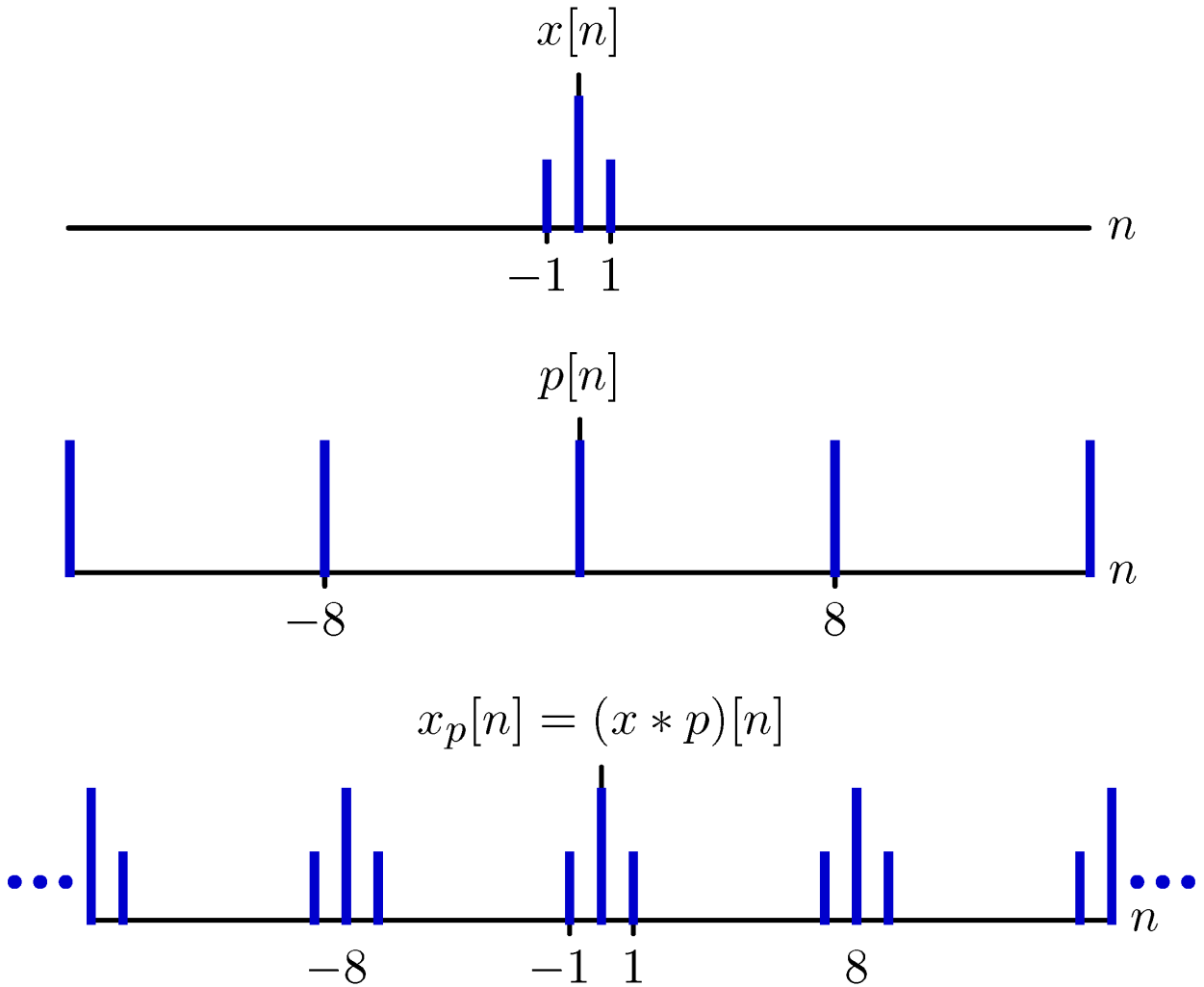
= multiplying by impulse train in frequency

$\rightarrow$ sampling in frequency



Relation Between DT Fourier Transform and Series

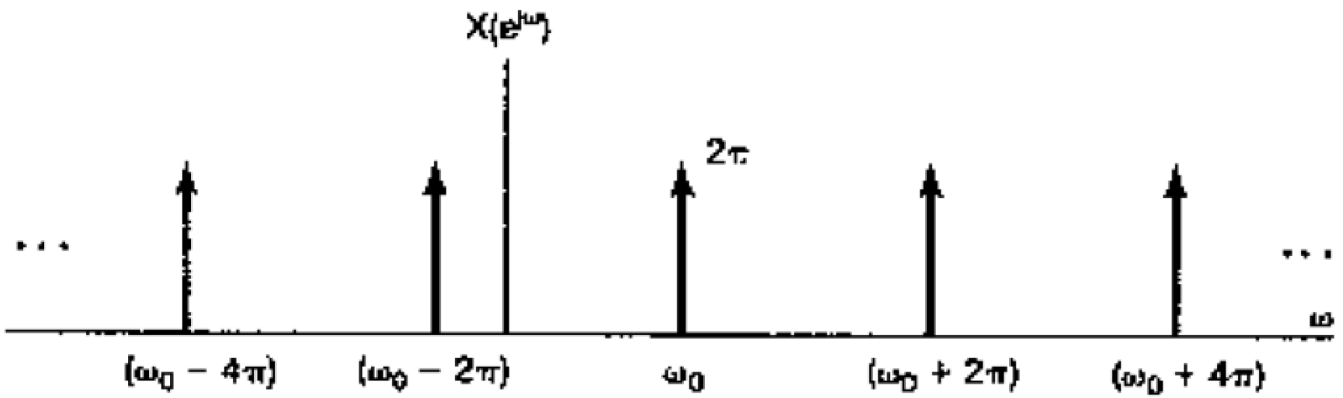
Periodic extension of a DT signal is equivalent to convolution of the signal with an impulse train.



DTFT of Periodic Signals

$$x[n] = e^{j\omega_0 n} \quad \Longleftrightarrow \quad X(e^{j\omega}) = ?$$

$$X(e^{j\omega}) = 2\pi \sum_{m=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi m)$$



DTFT of Periodic Signals

DTFT of Periodic Signals $x[n] = x[n+N]$

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n}, \omega_0 = \frac{2\pi}{N}$$

DTFS synthesis eq.

From the last page: $e^{jk\omega_0 n} \longleftrightarrow 2\pi \sum_{m=-\infty}^{\infty} \delta(\omega - k\omega_0 - 2\pi m)$

$$X(e^{j\omega}) = \sum_{k=<N>} a_k \left[2\pi \sum_{m=-\infty}^{\infty} \delta(\omega - k\omega_0 - 2\pi m) \right]$$

Linearity of DTFT

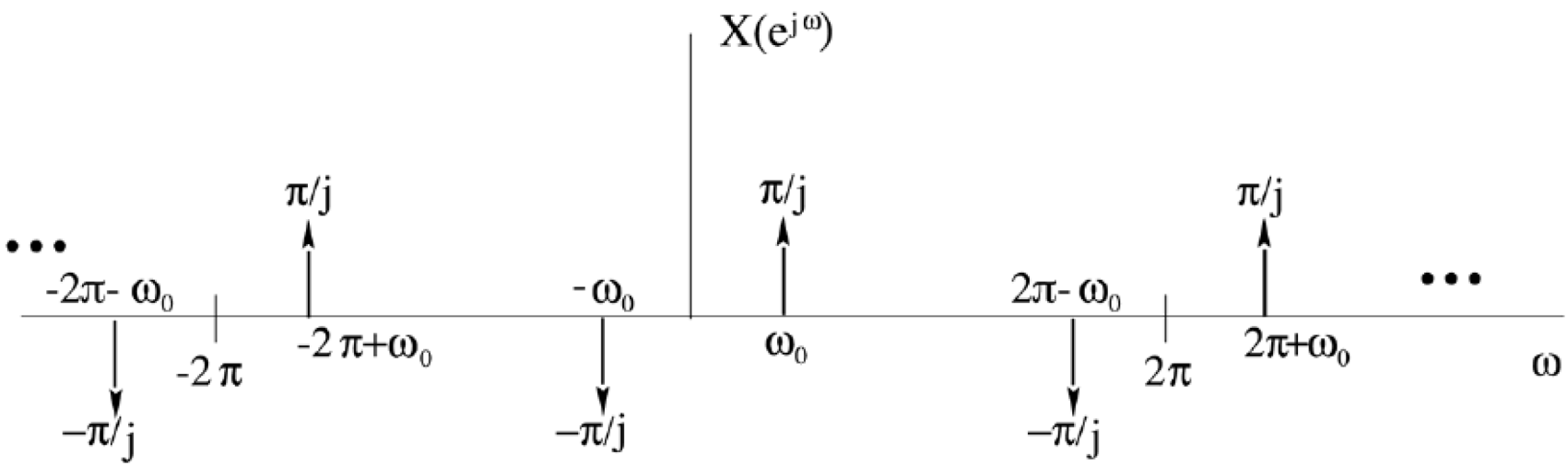
$$= 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$$

DTFT of Sine Signals

Ex. DT sine function

$$x[n] = \sin \omega_0 n = \frac{1}{2j} e^{j\omega_0 n} - \frac{1}{2j} e^{-j\omega_0 n}$$

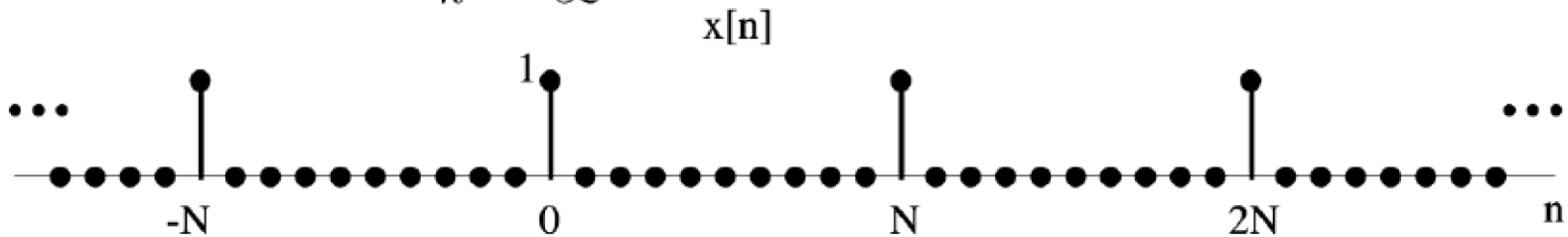
$$X(e^{j\omega}) = \frac{\pi}{j} \sum_{m=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi m) - \frac{\pi}{j} \sum_{m=-\infty}^{\infty} \delta(\omega + \omega_0 - 2\pi m)$$



DTFT of Periodic Signals

Ex. DT periodic impulse train

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN] \qquad \omega_0 = 2\pi/N$$

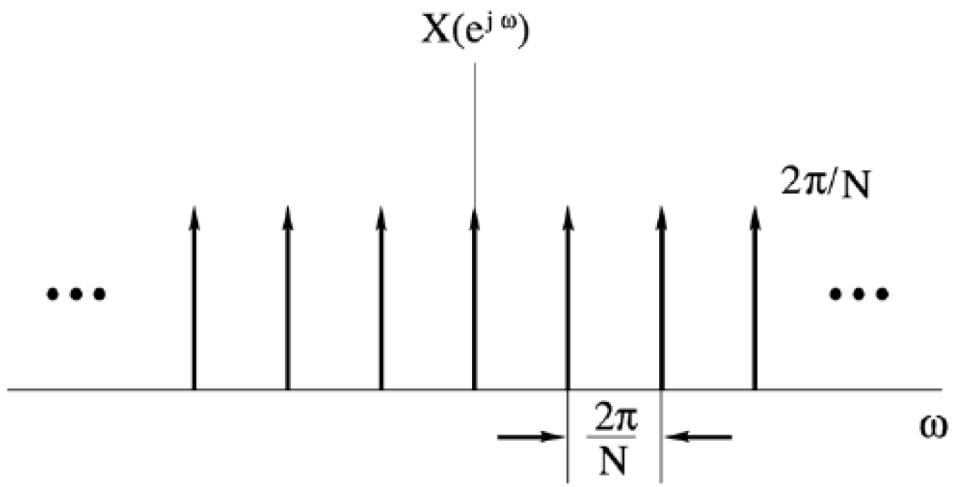


$$a_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n}$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} \underbrace{x[n]}_{=\delta[n]} e^{-jk\omega_0 n} = \frac{1}{N}$$

$$\Downarrow$$

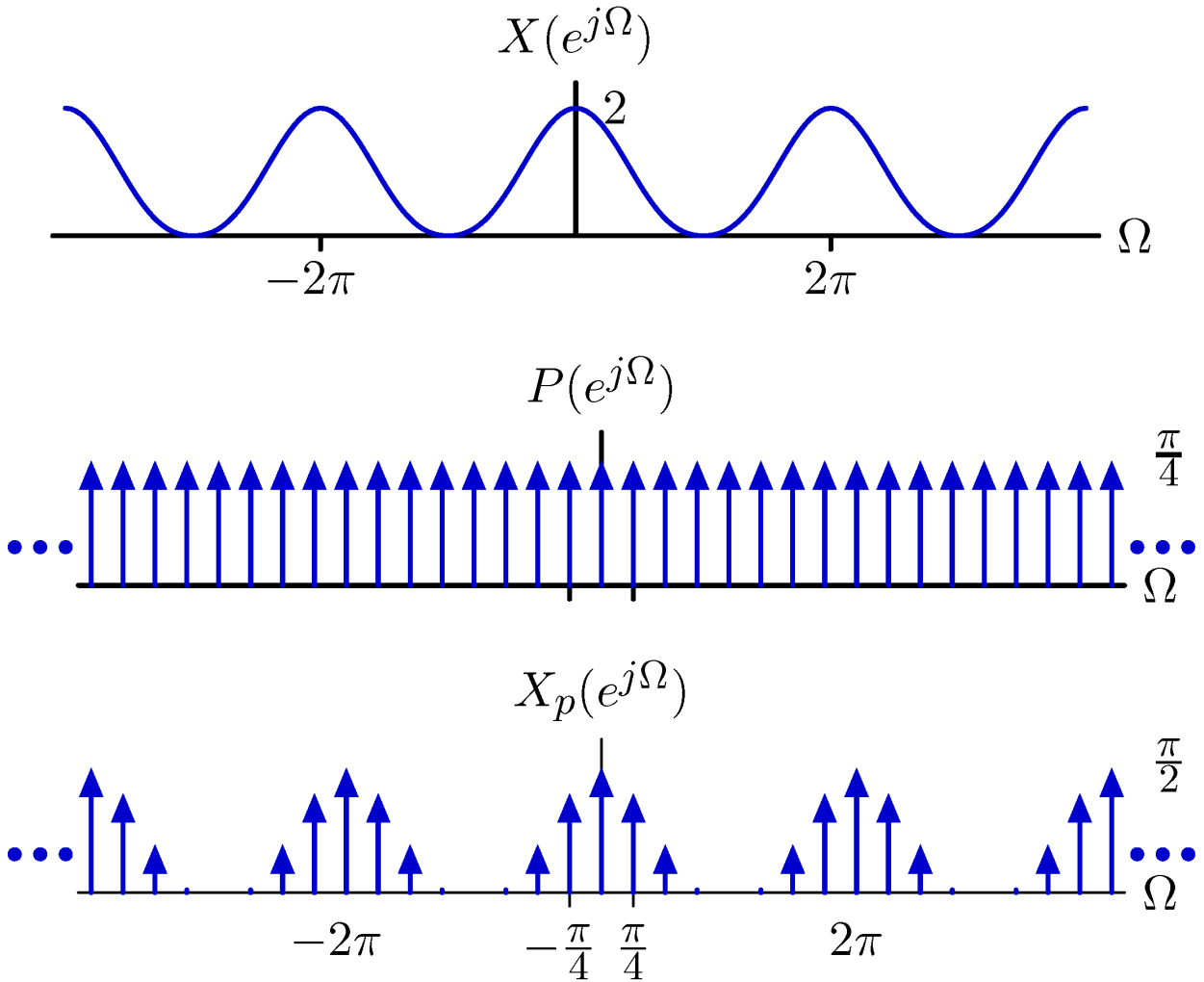
$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$$



— Also periodic impulse train – in the frequency domain!

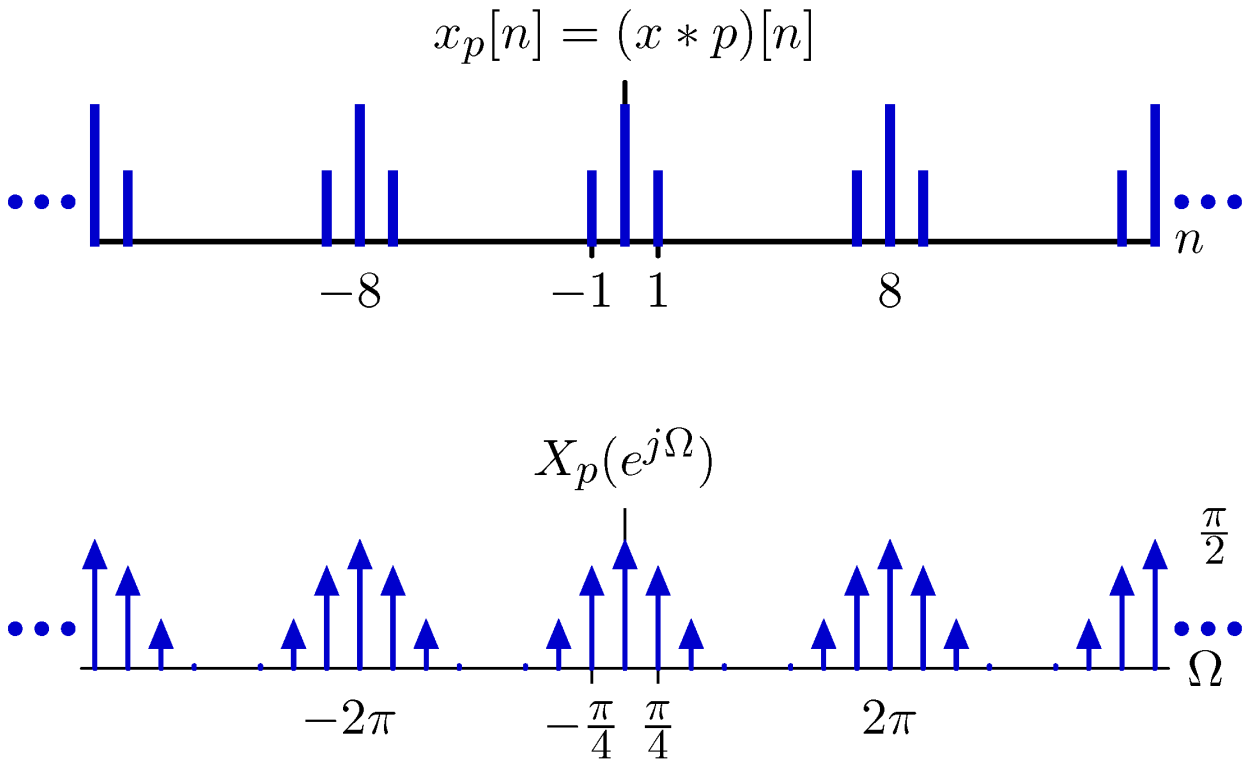
Relation Between DT Fourier Transform and Series

Convolution by an impulse train in time is equivalent to multiplication by an impulse train in frequency.



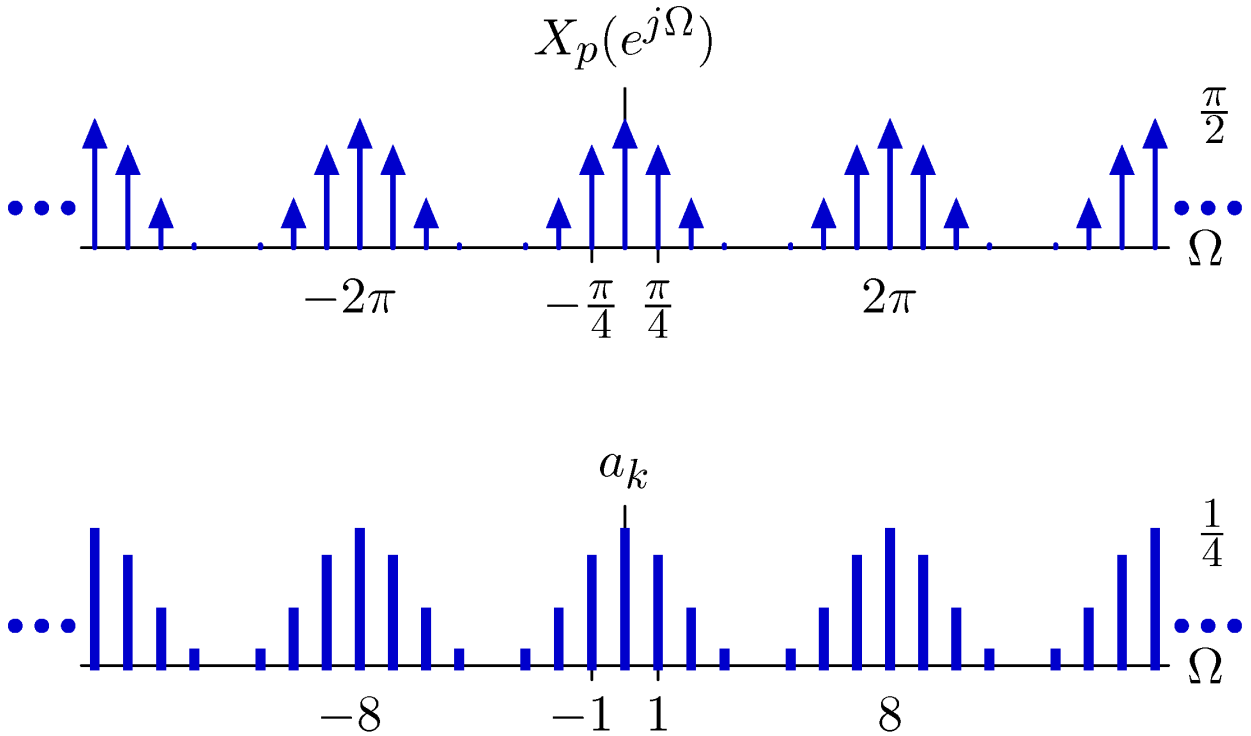
Relation Between DT Fourier Transform and Series

Periodic extension of a discrete signal ($x[n]$) results in a signal ($x_p[n]$) that is both periodic and discrete. Its transform ($X_p(e^{j\Omega})$) is also periodic and discrete.



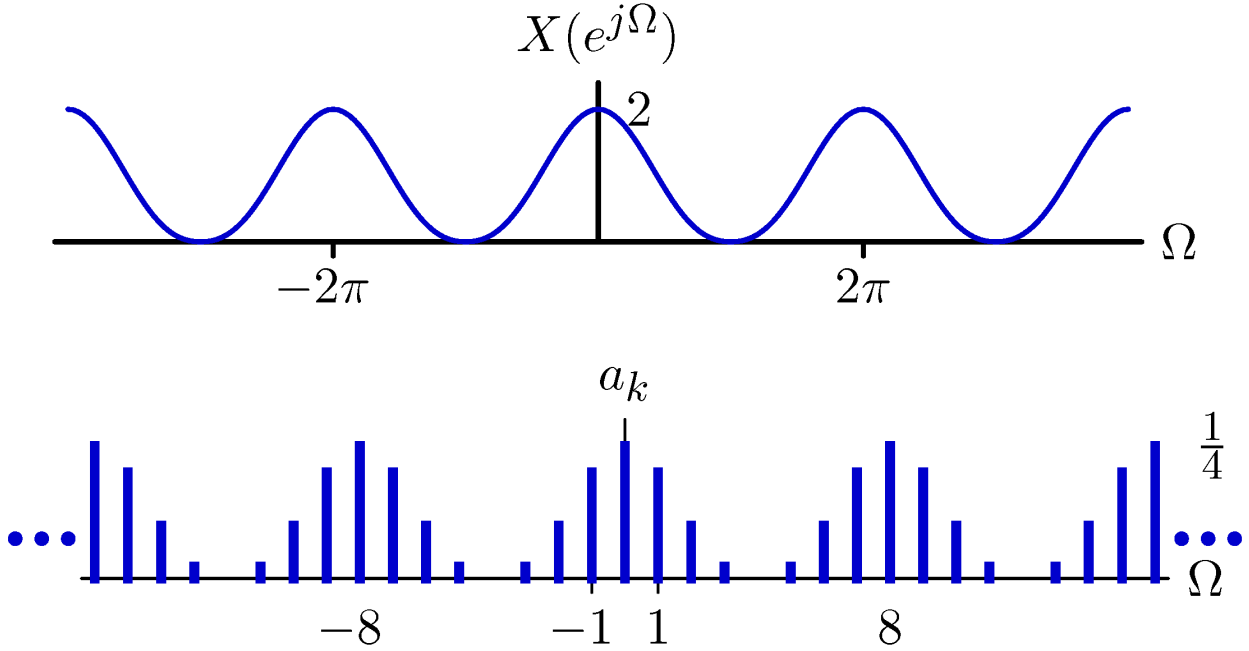
Relation Between DT Fourier Transform and Series

The weight of each impulse in the Fourier transform of a periodically extended function is 2π times the corresponding Fourier series coefficient.



Relation Between DT Fourier Transform and Series

The effect of periodic extension was to sample the frequency representation.



Relation between Fourier Transforms and Series

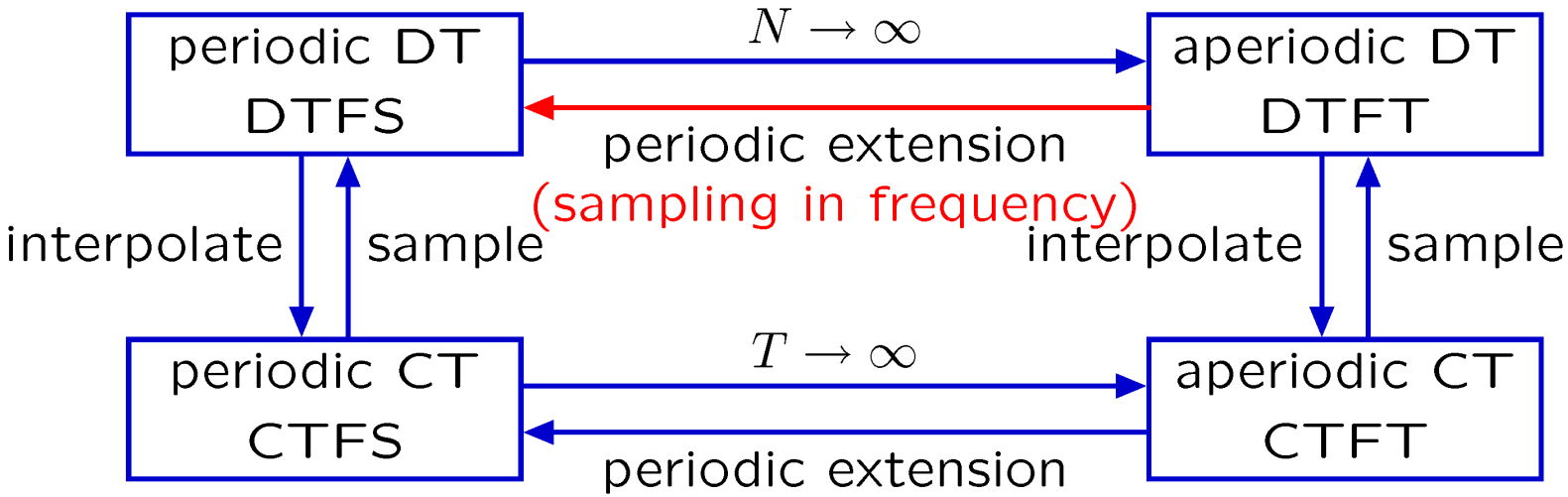
Periodic extension of a DT signal produces a discrete function of frequency.

Periodic extension

= convolving with impulse train in time

= multiplying by impulse train in frequency

$\rightarrow$ sampling in frequency



Duality

Fourier Transform is highly symmetric

CTFT: *Both* time and frequency are continuous and in general *aperiodic*

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Same except for these differences

$$\Downarrow$$

Suppose $f(\cdot)$ and $g(\cdot)$ are two functions related by

$$f(r) = \int_{-\infty}^{\infty} g(\tau)e^{-jr\tau}d\tau$$

Then

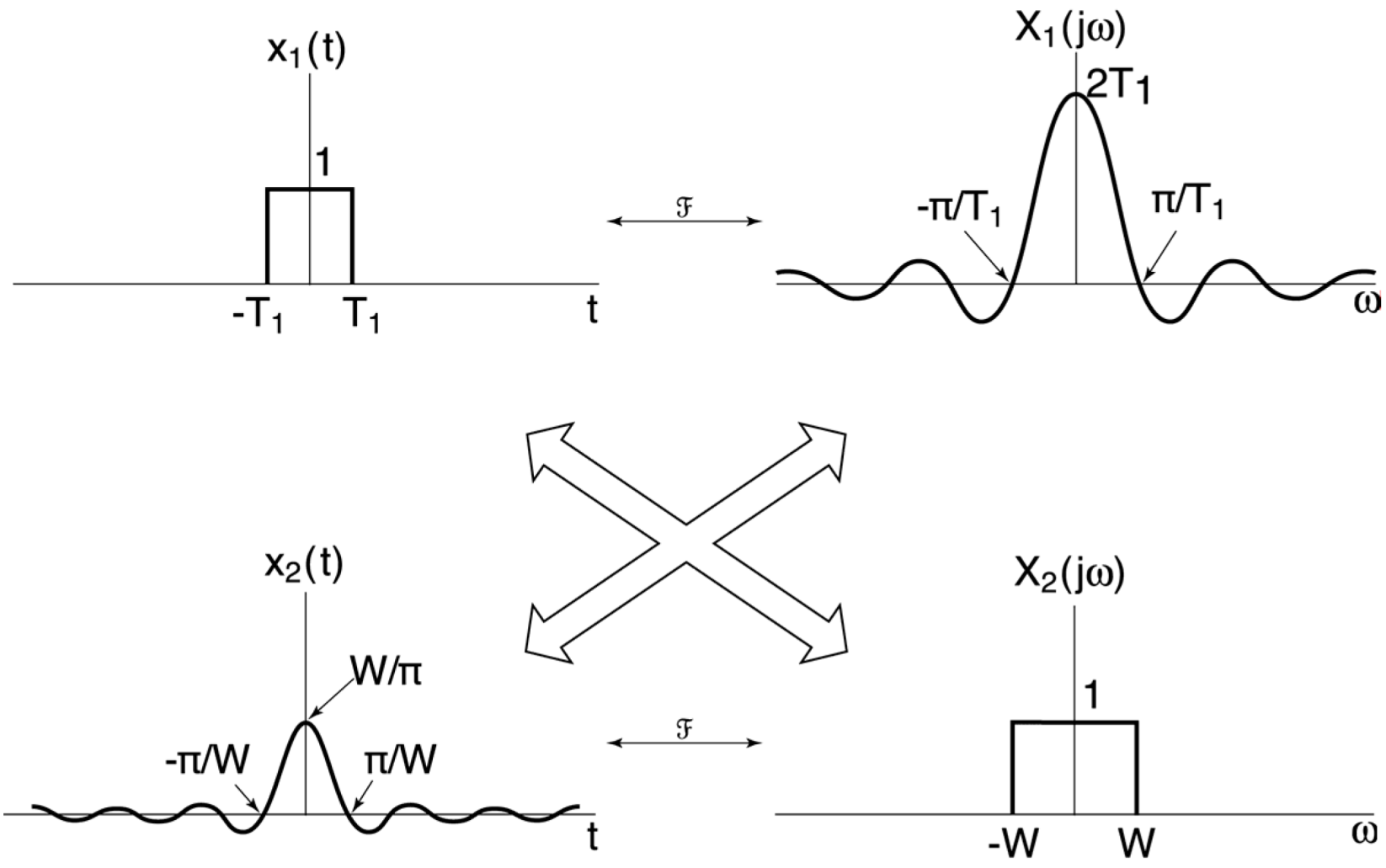
Let $\tau = t$ and $r = \omega$: $\quad x_1(t) = g(t) \longleftrightarrow X_1(j\omega) = f(\omega)$

Let $\tau = -\omega$ and $r = t$: $\quad x_2(t) = f(t) \longleftrightarrow X_2(j\omega) = 2\pi g(-\omega)$

Duality

Example of CTFT duality

Square pulse in either time or frequency domain



Duality

DTFS

Discrete & periodic in time $\longleftrightarrow$ *Periodic & discrete* in frequency

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = x[n+N], \quad \omega_0 = \frac{2\pi}{N}$$

$$a_k = \frac{1}{N} \sum_{k=<N>} x[n] e^{-jk\omega_0 n} = a_{k+N}$$

Duality in DTFS

Suppose $f[\cdot]$ and $g[\cdot]$ are two functions related by

$$f[m] = \frac{1}{N} \sum_{r=<N>} g[r] e^{-jr\omega_0 m}$$

$$\Rightarrow g[r] = \sum_{m=<N>} f[m] e^{jr\omega_0 m}$$

Then

Let $m = n$ and $r = -k$: $\quad x_1[n] = f[n] \longleftrightarrow a_k = \frac{1}{N} g[-k]$

Let $r = n$ and $m = k$: $\quad x_2[n] = g[n] \longleftrightarrow a_k = f[k]$

Duality

Duality between CTFS and DTFT

CTFS

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = x(t+T), \quad \omega_0 = \frac{2\pi}{T}$$

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt$$

Periodic in time $\leftrightarrow$ *Discrete* in frequency

DTFT

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = X\left(e^{j(\omega+2\pi)}\right)$$

Discrete in time $\leftrightarrow$ *Periodic* in frequency

CTFS-DTFT Duality

Suppose $f(\cdot)$ is a CT signal and $g[\cdot]$ a DT sequence related by

$$f(\tau) = \sum_{m=-\infty}^{+\infty} g[m] e^{jm\tau} = f(\tau + 2\pi)$$

Then

$$x(t) = f(t) \longleftrightarrow a_k = g[k]$$
(periodic with period 2π)

$$x[n] = g[n] \longleftrightarrow X(e^{j\omega}) = f(-\omega)$$

Parseval Relation

Magnitude and Phase of FT, and Parseval Relation

CT:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

$$X(j\omega) = |X(j\omega)|e^{j\angle X(j\omega)}$$

Parseval Relation:

$$\int_{-\infty}^{\infty}|x(t)|^2dt = \int_{-\infty}^{\infty}\frac{1}{2\pi}\underbrace{|X(j\omega)|^2}d\omega$$

Energy density in ω

DT:

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

$$X(e^{j\omega}) = |X(e^{j\omega})|e^{j\angle X(e^{j\omega})}$$

Parseval Relation:

$$\sum_{n=-\infty}^{\infty}|x[n]|^2 = \int_{2\pi}\frac{1}{2\pi}|X(e^{j\omega})|^2d\omega$$

Assignments

- Reading Assignment: Ch. 5